



Product Information & **MORE**

Customer Solutions

Amadeus PNR Data Feed

Version 1.5

Functional Marketing Specification

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Associated Documents

Document Name	Description
PNR Data Feed PO	Product overview of the PNR Data Feed
PNR Data Feed Presentation	Product Presentation of PNR Data Feed
AO2 Spec V1.11 (Feed V8.3)	Amadeus Internal technical document

Warning: This document may contain future functional developments or features applicable to the distinct system configurations of the receiving system and therefore technically not applicable in a generic manner to all airline systems.

Executive Summary

Clients may need to have visibility of the business generated by agencies and the activity of the PNRs, despite the fact they have sub-contracted reservation or inventory systems to Amadeus. They may also need to carry out revenue integrity, revenue management or operational process on PNRs which requires them to get PNR data when changes are made to bookings.

To help fulfil some of these needs Amadeus provides the PNR Data Feed product. On its side, the customer will then process the received PNRs and use the information provided.

This marketing document goes into greater functional detail than the Product Overview. It is designed to help customers understand how the product works and what they can expect if they reach a commercial agreement with Amadeus.

This product is based on the information contained in the SBR server (replication of the 1A central system PNR database). The system will elect PNRs to be sent to the client, based on a list of criteria selected by the airline. Any update of one of these elements can trigger the PNR to be sent. The PNR election is also based on the presence of a target airline's flight within the PNR or on the ownership of the PNR. Selected PNRs are either sent immediately to the client, or saved in a file and sent later via FTP on a regular basis (frequency chosen by the airline). The PNR information sent to the client can be filtered to convey only the relevant information. The information transmitted takes the form of SBRRES EDIFACT or XML messages. Information in this document relates to V8.3 of the PNR Data Feed.

Introduction

Problem Statement

Description

Clients need to follow up closely their business; the generated needs might be to:

- Leverage their Customer Relationship Management (CRM) system
- Better evaluate the business generated by their agencies
- Carry out revenue integrity, revenue management, strategic planning or operational activities based on PNR Data.
- Follow precisely the life cycle of the Passenger Name Record (PNR) from the booking, through the ticketing and after the departure

Amadeus (1A) could help fulfilling these needs through the PNR data feed product. This product provides the option to track the evolution of PNRs based on certain criteria and on a chosen frequency. By this product 1A provides the data basis for further analysis to be handled on customer's side.

Context

The system will be used as a data provider by the client's decisional systems (data warehouse, CRM...).

Another usage of the PNR data feed is to be the data source to perform some monitoring on PNR activities (agency monitoring, agency blacklist, etc...).

The PNR database of the SBR server is the data source of the system.

The information will be transmitted in EDIFACT format (using message SBRRES), or XML format.

Product Overview

Description

The PNR data feed aims at sending relevant PNR data, on the right frequency to the customer's system. The data is extracted from the SBR server database.

The customer chooses amongst several implementation options. The choice is made on the following items:

- PNR eligibility option(s): At each EOT, we check if the PNR can be part of a PNR Data Feed. If yes, the PNR is eligible for this feed. For example, a reason to be eligible for a 6X feed would be to have a 6X segment in the PNR.
- PNR view: Depending on the eligibility options and on what the client is authorized to see in the PNR, restrictions are applied on the elements that can be sent through the PNR Data Feed. The result of those restrictions is called the view.
- Triggering condition(s): For each eligible PNR, we check if the PNR view matches some rules defined by clients. If yes, the PNR will be sent through the PNR Data Feed. A triggering condition may be for example the addition/modification/deletion of an element in a PNR.

- Output selection: Clients can choose to receive only a set of elements from the PNR. For example they can choose to receive everything except the remarks.
- Data Feed mode (PNR image sent just after EOT or on a regular time basis)
- Data encoding format (EDIFACT or XML)

Depending on this configuration the PNR data feed will:
Watch the eligible PNRs in order to send one or several PNR images each time a trigger condition has been reached. The images are sent either immediately after the trigger condition has been verified (interactive online mode), or on a regular basis (deferred mode).

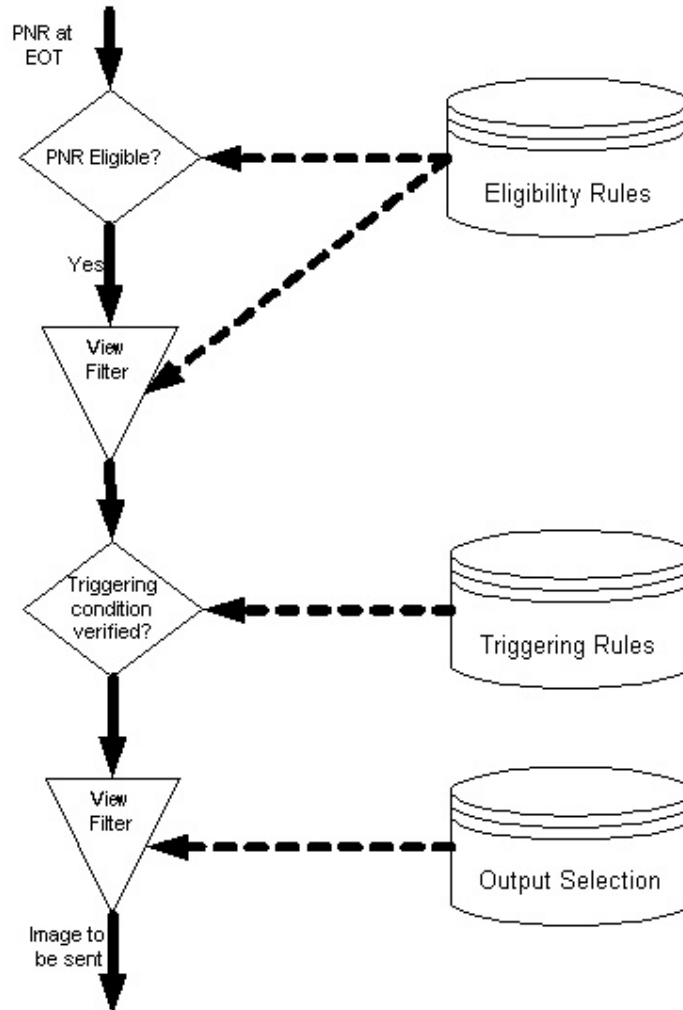


Figure 1: Eligibility - Trigger - View

Context

N/A

Prerequisites

A private network link is set up between 1A and the recipient system.

Limitations

The SBR Server - Central system synchronization is out of scope of this document.
How the receiving system decodes and processes the data is out of scope.

Functional Specifications

Eligibility Check

Definition

For each PNR present on the Reservation System and after each EOT, we check if the PNR matches the eligibility conditions to be part of a PNR Data Feed. If it does, the PNR is “eligible”.

A PNR being eligible is not sufficient for a PNR to be sent to the customer: The view restrictions (cf. 0 View) have to be applied and the triggering conditions (cf. 0 Trigger) have to be checked.

The eligibility criteria differ depending on each implementation of the PNR data feed. They are chosen by the customer, depending on which set of PNRs they are interested in.

Eligibility conditions

To become eligible for an airline's feed, a PNR must satisfy one of the following conditions (to be chosen by clients)

Eligibility by segments¹

- Presence of a carrier's segment in the PNR
- Presence of a segment part of a given list of flight numbers (Airline code + Flight number)

Eligibility by office

- Owner² office is part of a given list of Office IDs
- Owner office matches a certain pattern (ex.: ***6X0***)

Note for Marketing PNRs and eligibility by segments:

The segment can be normal air - prime or marketing - segments (including Passive and Ghosts), Open or Informational segments.

Note for Operating PNRs and eligibility by segments:

Operating PNRs are dedicated to a specific operating airline. When checking if an operating PNR is eligible for a PNR Data Feed, informational air segments are not taken into account.

Note for Single PNRs and eligibility by segments:

¹ It can be normal air segments (including Passive and Ghosts), Open segments or Informational segments.

² The Owner (or the Responsible) of a PNR is the first field in the RP line. The second field in the RP line is the queuing office, and there is no eligibility rule based on the queuing office. Note that in the case of OA PNRs, those fields may not contain Amadeus Office ID.

If 7X has subscribed to Single PNRs, a PNR containing a 6X segment operated by 7X will be eligible for a PNR Data Feed where eligibility is “PNR contains a 6X segment” and for a PNR Data Feed where eligibility is “PNR contains a 7X segment”.

Example 1 – Simple eligibility based on presence of a 6X segment

Assumption: A PNR is eligible for the PNR Data feed to 6X when it contains a 6X segment
The PNR below is eligible for 6X

```

--- RLR ---
RP/MUC1A0701/MUC1A0701          LS/SU   1AUG05/1133Z   X8M9VA
1. TEST/A
2  6X3075 C 23DEC 5 LHRNCE HK1          1005 1305   *1A/E*
3  AP 0123456789-H
4  TK 0K01AUG/MUC1A0701
  
```

Example 2 – Simple eligibility for a marketing PNR (Codeshare)

Assumption: A PNR is eligible for the 6X PNR Data Feed when it contains a 6X segment
The PNR below is *not* eligible for 6X, only the Operating PNR will be transmitted to the 6X ADS database.

```

--- RLR ---
RP/MUC1A0701/MUC1A0701          LS/SU   1AUG05/1133Z   X8M9VA
1. TEST/A
2  8X7372 M 28MAY 1 LHRSYD HK1          4  1215 1850+1 *1A/E*
    OPERATED BY 6X AIRWAYS
3  AP 0123456789-H
4  TK 0K01AUG/MUC1A0701
  
```

However, after the addition of a 6X prime or marketing segment, this PNR (X8M9VA) will become eligible.

Example 3 – Simple eligibility for marketing and operating carrier (Codeshare segment)

Assumption: A PNR is eligible for the PNR Data feed to respectively 6X/7X/8X when it contains respectively a 6X/7X/8X segment

Marketing PNR:

```

--- RLR ---
RP/MUC1A0701/MUC1A0701          BM/PR   5APR06/1605Z   ZEDLTJ
1. DUPONT/MARC
2  6X 798 Y 20JUL 4 LHRHEL HK1          1  1815 2310   *1A/E*      Prime
3  8X 831 Y 21JUL 5 HELLHR HK1          2  0800 0910   *1A/E*      Prime
4  6X6080 Y 22JUL 6 LHRHEL HK1          1  1610 2105   *1A/E*      Operated by 8X
5  8X5909 Y 23JUL 7 HELLHR HK1          2  1815 1925   *1A/E*      Operated by 6X
6  6X7312 Y 24JUL 1 LHRSYD HK1          4  2215 0605+2 *1A/E*      Operated by 7X
7  7X 005 Y 30JUL 7 SYDSIN HK1          1  1550 2140      E*        Prime
8  7X 319 Y 04AUG 5 SINLHR HK1          1  2325 0555+1 *1A/E*      Operated by 6X
9  AP MUC - AMADEUS DEFAULT OFFICE - A
10 TK 0K05APR/MUC1A0701
  
```

```
>RLO
RP/MUC1A0701/MUC1A0701          BM/PR   5APR06/1605Z   ZEDLTJ
8X6X/ZEDLTL      6X8X/ZEDLTO      6X7X/ZEDLTT      7X6X/ZEDLTM
```

This PNR is eligible for 6X, 7X and 8X

Corresponding operating PNR 6X8X:

```
** OPERATING PNR **
--- RLR ---
```

```
RP/MUC6X/MUC6XZEDLTJ/MUC1A0701/99      5APR06/1605Z   ZEDLTO
1.DUPONT/MARC
2  6X 798 Y 20JUL 4 LHRHEL HK1           1815 2310   *I*
3  8X 831 Y 21JUL 5 HELLHR HK1           0800 0910   *I*
4  8X 840 Y 22JUL 6 LHRHEL HK1           1  1610 2105   E*
   MARKETING BY 6X 6080 Y
5  8X5909 Y 23JUL 7 HELLHR HK1           1815 1925   *I*
6  6X7312 Y 24JUL 1 LHRSYD HK1           2215 0605+2 *I*
7  7X 005 Y 30JUL 7 SYDSIN HK1           1550 2140   *I*
8  7X 319 Y 04AUG 5 SINLHR HK1           2325 0555+1 *I*
9 TK 0K05APR/MUC6X08X1
```

This PNR is an operating PNR for 8X, and is eligible for 8X only as it contains a 8X non informational segment. 6X and 7X segments are informational; hence they are not taken into account for the eligibility of the operating PNR.

Example 4 – Simple eligibility based on the owner office ID

Assumption: A PNR is eligible for the PNR Data feed to 6X when it is owned by a 6X0 ATO/CTO.

The PNR is eligible as soon as it is considered as being under the responsibility of an Amadeus office belonging to the customer. The office of responsibility has a specific corporate code and qualifier. For instance for 6X, the ATO/CTO office will verify the following syntax: ***6X0***, 6X0 being the corporate code (6X) and qualifier (0).

This PNR is eligible for 6X because it is owned by a 6X office (here MUC6X0701) although it does not contain any 6X segment.

```
--- RLR ---
RP/MUC6X0701/MUC8X0101          BM/PR   30NOV05/1621Z   2XATBI
1.TEST/A
2  7X9501 Y 01DEC 4 SYDLHR HK1           1825 0635+1
3  7X9041 Y 04DEC 4 LHRSYD HK1           2005 0635+1
4 AP MUC - AMADEUS DEFAULT OFFICE - A
5 TK 0K30NOV/MUC1A0701
6 SSR 0THS 7X TEST
```

Note that even if one of the 7X segment has been added in the PNR by MUC8X0101, the PNR is not eligible for a feed to 8X (eligibility by segment or by owner office)

Example 5 – Simple eligibility based on flights

Assumption: A PNR is eligible for the PNR Data feed to 7X when it contains a segment on flight 7X9501 on 01DEC05 or 02DEC05.

Hence the following PNR is eligible.

```

--- RLR ---
RP/MUC6X0701/MUC6X0701          BM/PR  30NOV05/1621Z    2XATBI
1.TEST/A
2  7X9501 Y 01DEC 4 SYDLHR HK1      1825 0635+1
3 AP MUC - AMADEUS DEFAULT OFFICE - A
4 TK OK30NOV/MUC1A0701
5 SSR OTHS 7X TEST
  
```

But the following one is not eligible because it contains flight 7X 9501 on 03DEC05.

```

--- RLR ---
RP/MUC6X0701/MUC6X0701          BM/PR  30NOV05/1621Z    2XATBI
1.TEST/A
2  7X9501 Y 03DEC 4 SYDLHR HK1      1825 0635+1
3 AP MUC - AMADEUS DEFAULT OFFICE - A
4 TK OK30NOV/MUC1A0701
5 SSR OTHS 7X TEST
  
```

Example 6 – Simple eligibility based on flight numbers

Assumption: A PNR is eligible for the PNR Data feed to 7X when it contains a segment on flight 7X9501 (so regardless of the date).

Hence the following PNR is eligible:

```

--- RLR ---
RP/MUC6X0701/MUC6X0701          BM/PR  30NOV05/1621Z    2XATBI
1.TEST/A
2  7X9501 Y 01DEC 4 SYDLHR HK1      1825 0635+1
3 AP MUC - AMADEUS DEFAULT OFFICE - A
4 TK OK30NOV/MUC1A0701
5 SSR OTHS 7X TEST
  
```

Example 7 – Eligibility for Single PNRs

Assumption: For this example, we consider that 7X have subscribed to Single PNRs.

6X is receiving a PNR Data Feed where eligibility is: PNRs containing a 6X segment.

7X is receiving a PNR Data Feed where eligibility is: PNRs containing a 7X segment.

The following single PNR (retrieved in a 7X office) is eligible for 6X (because it contains a 6X marketing segment) and also for 7X (because it contains a 7X operating segment)

```

RP/MUC6X0701/MUC6X0701          OC/SU  22JUN05/0917Z    YMRBBB
1.TEST/A
2  6X 321 F 13JUL 3 LGWAMS HK1      S  1000 1100    *1A/E*
   OPERATED BY 7X345 B
  
```

```
3 AP MNL - ORIENT EXPRESS - A
4 TK OK22JUN/MUC6X0701
```

Multi-eligibility

Different simple eligibility conditions can be combined.

Multi-eligibility can be set with:

- Simple Eligibility Condition 1 **or** Simple Eligibility Condition 2

Note: If multiple eligibility conditions match simultaneously, eligibility by OfficeID has the priority.

Example 8 – Multi-eligibility OR

Assumption: A PNR is eligible for the PNR Data feed to 6X either when it contains a 6X segment, or when it is owned by a 6X0 ATO/CTO.

In this case, each PNR which verifies at least one of these eligibility criteria is eligible. So both PNRs below are eligible for 6X feed.

```
--- RLR ---
RP/MUC1A0701/MUC1A0701          BM/PR  2DEC05/0814Z  2XBH5X
1.TEST/B
2  6X7311 Y 02DEC 5 SYDLHR HK1      1815 0625+1 *1A/
3 AP MUC - AMADEUS DEFAULT OFFICE - A
4 TK OK02DEC/MUC1A0701

--- RLR ---
RP/NCE6X0001/NCE6X0001          BM/PR  2DEC05/0819Z  2XBH6B
1.TEST/C
2  7X6211 Y 02DEC 5 NCEORY HK1  1045 2 1105 1230  *1A/E*
3 AP MUC - AMADEUS DEFAULT OFFICE - A
4 TK OK02DEC/MUC1A0701
```

Example 9 – Multi-eligibility OR

The multi-eligibility can be used for a common PNR Data Feed: For example 6X and 7X want to share one unique feed containing PNRs having a 6X or a 7X segment.

Assumption: A PNR is eligible for the PNR Data feed to 6X-7X when it contains a 6X or a 7X segment.

The 2 following PNRs are eligible for this Shared PNR Data Feed

```
--- RLR ---
RP/MUC1A0701/MUC1A0701          BM/PR  2DEC05/0814Z  2XBH5X
1.TEST/B
2  6X7311 Y 02DEC 5 SYDLHR HK1      1815 0625+1 *1A/
3 AP MUC - AMADEUS DEFAULT OFFICE - A
4 TK OK02DEC/MUC1A0701

--- RLR ---
```

RP/NCE6X0001/NCE6X0001	BM/PR	2DEC05/0819Z	2XBH6B
1. TEST/C			
2 7X6211 Y 02DEC 5 NCEORY HK1 1045 2 1105 1230 *1A/E*			
3 AP MUC - AMADEUS DEFAULT OFFICE - A			
4 TK 0K02DEC/MUC1A0701			

View

When a PNR is eligible for a PNR Data Feed, it is retrieved with a point of view corresponding to what the client is authorized to see. This restricted image of the PNR is called the view of the PNR.

The same PNR may have different views if it is eligible for different PNR Data Feed.

When building the view, security restrictions are applied first, and restrictions due to the type of eligibility after.

Security restrictions

The PNR data feed runs and access PNRs through a list of dedicated offices (having the following pattern: MUCxxxDFy where xxx is a corporate code/qualifier and y is the number of the feed in the case of several feeds for one client) and using a chosen duty code (by default RC)³.

These offices have some access rights. If an element is visible in at least one of these offices, it will be available in the view. But if an element of a PNR appears as "RESTRICTED" when retrieved in each security office used for a specific PNR Data Feed, it will never be part of the view for this specific PNR Data Feed.

Security restrictions for code share and inventory information

At segment level, the following data may be present in the PNR:

- Segment details (carrier code, flight number, booking class, date/time...)
 - (In case of code share, these data are related to the segment's carrier code)
- Code share details
 - o Partner carrier code, flight number
 - o Type of Code share agreement (free flow, block space, single)
 - o Partner booking class
- Inventory details
 - o Cabin code
 - o Sub-class
 - o Sub class classification
- Overbooking details and reason
- Yield, Bid-price

1. The following data are sent to any PNR Data Feed customer:

- Segment details (carrier code, flight number, booking class, date/time...)

³ The corporate code/qualifier and the duty code to use are chosen by clients in the implementation document.

- Code share details
 - o Partner carrier code, flight number
 - 2. The following data are sent only to both carriers involved in the code share agreement (the marketing and the operating carriers):
 - Code share details
 - o Type of Code share agreement (free flow, block space, single)
 - o Partner booking class
- This applies to Single PNR, Marketing PNR or Operating PNR.

3. The following data are sent only to the carrier's segments (not to the Code share partner):
 - Inventory details
 - o Cabin code
 - o Sub-class
 - o Sub class classification
 - Overbooking details and reason
 - Yield, Bid-price
- This applies to Single PNR, Marketing PNR, Operating PNR or prime PNR.

Code share and inventory information are sent as described above provided that security office(s) attached to the PNR Data feed customer are defined in the PNR Data Feed publication. If no security office is defined, these information are not sent in the PNR Data Feed.

Eligibility restrictions

Moreover, the view of certain elements is restricted depending on their segment association or corporate association (airline code). For instance, in a PNR sent to an airline customer, the view does not contain any elements that are explicitly associated to another airline segment only.

In case of multi eligibility, the view is the union of the different views corresponding to eligibility.

Depending on the reason why the PNR is eligible, security restrictions are applied as follows:

Eligibility by segment (for 6X)⁴:

- Excluding elements associated to flown segments (default view):

- 1-The full air itinerary is present (even flown segments)
- 2-Automated Non air segments (Hotel, Car) are not present.
- 3-Manual Auxiliary segments are present if they have 6X or 1A as carrier code
- 4-All the non segment associated elements are present (except FT, FY, MCO,SVC elements)
- 5-All the 6X and YY elements are present (except FT, FY, MCO,SVC elements)
- 6-All the 6X eligible segment associated elements are present (except if the 6X eligible segment is flown) (except FT, FY elements)
- 7-All the elements only associated to non 6X segments are not present

⁴ The receiver must choose in its implementation document to receive or not elements associated to flown segments.

8- MCO and SVC elements are present if the eligible airline (6X) is equals to the element carrier code

9-FT, FY elements and TST are present if:

- When FV element exists in the PNR:
The eligible airline (6X) is equals to the validating carrier (airline specified in FV element)
- When FV element does not exist in the PNR:
The PNR owner is an ATO/CTO of the eligible airline (6X). (Eligible airline ATO/CTO is such as office id corporate code = eligible airline and corporate qualifier is 0. (example LON6X0100 ATO/CTO of 6X)

Remark: if several FV elements exist in the PNR with different validating carriers, FT, FY elements and TST will be sent in the PNR data feed of the different airlines.

- Including elements associated to flown segments:

Same as above for cases 1-2-3-4-5-7-8-9

For case 6: All the 6X eligible segment associated elements are present even if the 6X eligible segment is flown.

Eligibility by segment for single Code share segments (6X*/ 7X):

Data sent to marketing carrier:

Same as above for cases 1-2-3-4-5-6-7-8-9

Remark for case 5: The elements associated to the marketing carrier (6X) and YY are sent.

Data sent to operating carrier:

Same as above for cases 1-2-3-4-6-7-8-9

In addition:

The elements associated to the marketing carrier (6X), the operating carrier (7X) and YY are sent.

Eligibility by office:

No restriction is applied

Specific eligibility rule for SSR OTHS/ADTK 1A or ZZ:

Even if in the PNR display it appears as an SSR OTHS/ADTK xx where xx is another GDS/Airline reservation system code, if this SSR OTHS/ADTK was entered as 1A in the PNR. It will be sent through the feed.

An incoming SSR OTHS/ADTK xx received through a TTY message where xx is not YY, 1A or 6X will not be part of the 6X PNR data feed view and thus never be sent through the feed. When an agent adds an SSR OTHS/ADTK ZZ or a SSR OTHS/ADTK 1A in Amadeus, it will be part of the PNR data feed view and then sent is chosen as an output, even if it is changed to SSR OTHS xx where xx is an other GDS code.

Examples:

Example 10 – View restriction mechanism. Multiple security offices

Assumption: A PNR is eligible for the PNR Data feed when it contains a 6X or a 7X segment. The offices used for the security of this PNR Data Feed are **MUC6X0DF0** and **MUC7X0DF0**.

```

--- RLR ---
RP/NCE6X0001/NCE6X0001          AA/GS  18SEP05/1515Z  X80DQL

1.TEST/A
2  6X6227 Y 20SEP 2 NCEORY HK1 1745 2 1805 1930  *1A/E*
3  7X 421 J 21SEP 3 SYDMEL HK1          0930 1055  *1A/E*
4 AP FRA-0123456789
6 TK OK03AUG/NCE6X0001
7 RC NCE8X0001-W,***8X****/THIS IS A 8X CONFIDENTIAL REMARK      (not visible in the
display and not present in the view)
8 RC NCE7X0001-W,***7X****/THIS IS A 7X CONFIDENTIAL REMARK      (visible in the display
and present in the view)
9 RC NCE6X0001-W,***6X****/THIS IS A 6X CONFIDENTIAL REMARK      (visible in the display
and present in the view)

```

Example 11 – View restriction mechanism. Eligibility by segment

Assumption: A PNR is eligible for the PNR Data feed to 6X when it contains a 6X segment. The office ID used for the security of this PNR Data Feed is MUC6X0DF0. 6X has chosen the view **including** elements associated to flown segments.

A PNR is eligible for the PNR Data feed to 7X when it contains a 7X segment. The office ID used for the security of this PNR Data Feed is MUC7X0DF0. 7X has chosen a view **excluding** elements associated to flown segments.

View construction for the PNR Data Feed to 6X:

The SSRs (element 8-9) will be part of the view because they are associated to 6X segments (even flown).

The remark (element 10) will not be part of the view as it is associated to a 7x segment.

The RC (elements 11) will be part of the view as the office ID used for the PNR Data Feed is a 6X ATO/CTO

```

--- RLR ---
RP/NCE6X0001/NCE6X0001          AA/GS  18SEP05/1515Z  X80DQL

1.TEST/A
2  6X 016 F 06SEP 1 SYDLHR HK1          FLWN
3  7X6220 Y 13SEP 2 ORYNCE              FLWN
4  6X6227 Y 20SEP 2 NCEORY HK1 1745 2 1805 1930  *1A/E*
5  7X 421 J 21SEP 3 SYDMEL HK1          0930 1055  *1A/E*
6 AP FRA-0123456789
7 TK OK03AUG/NCE6X0001
8 SSR VGML 6X HK1/S2              (Visible in the display and present in the view)
9 SSR OTHS 6X-This is free text    (Visible in the display and present in the view)
10 RM this is a remark/S3          (Visible in the display but not present in the view)
11 RC NCE6X0001-W,***6X****/THIS IS A 6X CONFIDENTIAL REMARK      (visible in the display
and present in the view)

```

View construction for the PNR Data Feed to 7X:

The same PNR is retrieved in a 7X office for the 7X PNR Data Feed. The confidential remark is hidden and specific 6X SSR is not sent. The confidential remark is not shown because of the security restriction mechanism. The SSRs (element 8-9) and the confidential remark (element 11) are not sent because of the eligibility/security restrictions. The remark (element 10) is not sent because it is associated to a flown segment.

```

--- RLR ---
RP/NCE6X0001/NCE6X0001          AA/GS  18SEP05/1515Z  X80DQL
1. TEST/A
2  6X 016 F 06SEP 1 SYDLHR HK1      FLWN
3  7X6220 Y 13SEP 2 ORYNCE          FLWN
4  6X6227 Y 20SEP 2 NCEORY HK1 1745 2 1805 1930  *1A/E*
5  7X 421 J 21SEP 3 SYDMEL HK1      0930 1055  *1A/E*
6 AP FRA-0123456789
7 TK OK03AUG/NCE6X0001
8 SSR VGML 6X HK1/S2              (Visible in the display but not present in the view)
9 SSR OTHS 6X-This is free text    (Visible in the display but not present in the view)
10 RM this is a remark/S3          (Visible in the display but not present in the view)
11 RC RESTRICTED                  (Not visible in the display and not present in the view)

```

Example 12 – View restriction mechanism. Eligibility by flight number

Assumption: A PNR is eligible for the PNR Data feed to 6X when it contains a 6X 317 segment. The office ID used for the security of this PNR Data Feed is MUC6X0DF0.

The SSR OTHS 6X (element 6) will be part of the view since it is associated to 6X (therefore to 6X317 segment).

The SSR OTHS (element 8) will not be part of the view as it is associated to a non eligible segment (6X179 is a non eligible flight).

The SSR OTHS (element 7) will be part of the view as it is associated to an eligible segment.

```

--- RLR ---
RP/MUC1A0701/MUC1A0701          BM/PR  22JAN07/1039Z  2G6R45
1. TEST/ELIGIBILITY ON FLIGHT
2  6X 317 Y 22JAN 1 CDGLHR HK1      1 1510 1525  *1A/E*
3  6X 179 Y 22JAN 1 LHRJFK HK1      4 1815 2100  *1A/
4 AP MUC - AMADEUS DEFAULT OFFICE - A
5 TK OK22JAN/MUC1A0701
6 SSR OTHS 6X THIS IS A 6X ASSOCIATED ELEMENT    (Visible in the display and present in the view)
7 SSR OTHS 6X HN1 THIS IS A 6X317 ASSOC. ELEMENT/S2  (Visible in the display and present in the view)
8 SSR OTHS 6X HN1 THIS IS A 6X179 ASSOC. ELEMENT/S3  (Visible in the display but not present in the view)

```

Example 13 – Example to avoid confusions for SSR FQTV

To avoid confusions with SSR FQTV, it is important to note that the view restriction is based on the carrier code of the SSR, and not on the carrier code present in the free text of the SSR.

Assumption: A PNR is eligible for the PNR Data feed to 6X when it contains a 6X segment.

SSR FQTV **6X** HK/ 6X123456789/6 → This SSR is part of the 6X view (as carrier code of the SSR is 6X)
 SSR FQTV **6X** HK/ 7X123456789/6 → This SSR is part of 6X view (as carrier code of the SSR is 6X)
 SSR FQTV **7X** HK/ 6X123456789/6 → This SSR is not part of 6X view, even if the frequent flyer number is a 6X frequent flyer number (as carrier code of the SSR is 7X)

Example 14 – Example for MCO

To avoid confusions with MCO, it is important to note that the view restriction is based on the carrier code of the MCO, and not on the carrier code that may be present in sub-fare elements associated to the MCO.

Assumption: A PNR is eligible for the PNR Data feed to 6X when it contains a 6X segment.

MCO 6X *** 30MAR/ → This MCO is part of the 6X view (as carrier code of the MCO is 6X)

MCO 6X *** 30MAR/
 1 FP CASH → This MCO is part of the 6X view and not of the 7X view.
 2 FT IT87X1TOUR99

Example 15 – View restriction mechanism. Eligibility by Office

Assumption: A PNR is eligible for the PNR Data feed to 7X when it is owned by a 7X0 ATO/CTO. The office ID used for the security of this PNR Data Feed is MUC7X0DF0.

The SSR VGML associated to the 6X segment will be part of the view as the owner office for this PNR is a 7X ATO/CTO (No eligibility restrictions are applied). However, the RC remark won't be part of the view because of the security restrictions.

```

--- RLR ---
RP/NCE7X0001/NCE7X0001          AA/GS   3AUG05/1515Z   X80DQL
1. TEST/A
2  6X 016 F 20SEP 2 SYDLHR HK1      1  1620 0455+1 *1A/
3  7X 421 J 21SEP 3 SYDMEL HK1      0930 1055   *1A/E*
4 AP FRA-0123456789
5 TK OK03AUG/NCE6X0001
6 SSR VGML 6X HK1/S2              (Visible in the display and present in the view)
7 RC RESTRICTED                    (Not visible in the display and not present in
the view)
  
```

Example 6 – View for Single PNRs

Assumption: For this example, we consider that 7X have subscribed to Single PNRs. 6X is receiving a PNR Data Feed where eligibility is: PNRs containing a 6X segment. 7X is receiving a PNR Data Feed where eligibility is: PNRs containing a 7X segment.

Below is the original Single PNR:

```

--- RLR ---
RP/NCE6X0100/NCE6X0100          AA/GS   6APR06/0941Z   ZEDZWV
1. TEST/A
2  6X7372 Y 06APR 4 LHRSYD DK1  1115 1925  07APR      1 744
   OPERATED BY 7X032 Y
3 AP NCE 111222333 - ABG PMG DEMO OFF NCE - A
  
```

4 SSR OTHS 6X THIS IS A MKT SSR
5 SSR OTHS 7X THIS IS AN OPERATING SSR

This PNR is eligible for 6X and 7X.
The view of the PNR for 6X will contain elements 1-2-3-4.
The view of the PNR for 7X will contain elements 1-2-3-4-5.

Example 7 – View construction for a Multi-eligibility

In case of multi-eligibility, the view is the union of the views built for each simple eligibility⁵.
Assumption: A PNR is eligible for the PNR Data feed to 6X-7X when it contains a 6X segment or a 7X segment. The office ID used for the security of this PNR Data Feed is MUC6X0DF0.

In the following PNR, elements 6-7-8-9 are part of the view:
The SSR (element 6) will be part of the view (due to the 6X view), the SSR (element 7) will be part of the view (due to the 7X view).
The RC (elements 8) will be part of the view, but the RC (elements 9) will not be part of the view because of the security restrictions.

```

--- RLR ---
RP/NCE6X0001/NCE6X0001          AA/GS   3AUG05/1515Z   X80DQL
1. TEST/A
2  6X 016 F 20SEP 2 SYDLHR HK1      1  1620 0455+1 *1A/
3  7X 421 J 21SEP 3 SYDMEL HK1      0930 1055  *1A/E*
4  AP FRA-0123456789
5  TK OK03AUG/NCE6X0001
6  SSR VGML 6X HK1/S2                (Visible in the display and present in the view)
7  SSR VGML 7X HK1/S3                (Visible in the display and present in the view)
8  RC NCE6X0001-W,***6X****/THIS IS A 6X CONFIDENTIAL REMARK      (visible in the 6X
display and present in the view)
9  RC NCE7X0001-W,***7X****/THIS IS A 7X CONFIDENTIAL REMARK      (visible in the 7X
display but not present in the view)

```

Example 18 – Example to avoid confusions for SSR ADTK or OTHS 1A or ZZ

Assumption: For this example, we consider received SSR OTHS/ADTK 1A through a TTY message A PNR is eligible for the PNR Data feed to 6X when it contains a 6X segment.

Below is the original PNR:

```

--- RLR ---
RP/NCE6X0100/NCE6X0100          LS/SU   6APR06/1433Z   ZEDZWV
1. TEST/A
2  6X7372 Y 06APR 4 LHRSYD DK1  1115 1925  07APR      1 744

```

⁵ In the case of the union of the marketing and the operating views of a segment, the resulting view will be the operating one .

OPERATED BY 7X032 Y	
3	AP NCE 111222333 - ABG PMG DEMO OFF NCE - A
4	SSR OTHS 6X Free text
5	SSR ADTK 6X Free text
6	SSR OTHS 7X Free text
7	SSR ADTK 7X Free text
8	SSR OTHS 1A Free text (entered as ZZ by an agent)
9	SSR ADTK 1A Free text (entered as ZZ by an agent)

This PNR is eligible for 6X.

The view of the PNR for 6X will contain elements 1-2-3-4-5-8-9

For OA PNRs:

--- RLR ---	
RP/SWIG1G/SWIG1GKKK	LS/SU 6APR06/1433Z ZEDZWV
1. TEST/A	
2	6X7372 Y 06APR 4 LHRSDY DK1 1115 1925 07APR 1 744
OPERATED BY 7X032 Y	
3	AP NCE 111222333 - ABG PMG DEMO OFF NCE - A
4	SSR OTHS 6X Free text
5	SSR ADTK 6X Free text
6	SSR OTHS 7X Free text
7	SSR ADTK 7X Free text
8	SSR OTHS 1G Free text (entered as 1A via TTY)
9	SSR ADTK 1G Free text (entered as ZZ by an agent)
10	SSR OTHS 1G Free text (entered as 1G via TTY)

This PNR is eligible for 6X.

The view of the PNR for 6X will contain elements 1-2-3-4-5-8-9

Triggering conditions

Definition

The triggering conditions are a set of “complex rules” defined by clients to decide whether or not an eligible PNR will be part of the PNR Data Feed. These “complex rules” are applied on the PNR view previously built. If at least one of them is verified, the PNR will be part of the PNR Data Feed.

A basic triggering rule is made of an element (and optionally the value of an element for some elements detailed below) and an activity.

Basic triggering rules can be combined together with OR or AND to build the “complex rules” (cf Example)

Elements have to be chosen in 0 Annex 2: PNR Data elements choice and granularity.

The activity can be:

- Addition: a new element has been added to the view since last EOT.
- Modification: an element has been modified in the view since last EOT.
- Deletion: an element has been deleted from the view since last EOT.
- Presence: the feed is triggered only because of the presence of an element in the view after EOT⁶.

NB: the activity of a PNR is analyzed after each EOT, except in the case of a simple queuing (ie retrieving a PNR and queuing it without any update on the face of the PNR). However, if a PNR is retrieved, updated and end transacted using a queuing entry, it will be analyzed and it can trigger the PNR Data Feed.

The trigger could be defined up to the level of element value⁷ for the following elements:

- SSR (Special Service Request)
- SK (Keyword)
- OSI (Other Service Information)
- RM (General remark)

If the element value or set of values is not specified, the element is taken as a triggering element whatever its value.

The triggering rules can be combined. For example a triggering rule can be: trigger the feed if the PNR contains an SSR FQTV and if an SSR VGML is added.

In the case of eligibility by segment, updates on the PNR should not trigger a feed for an airline that has **only flown** segments in the PNR.

In the case of eligibility by office, updates on a PNR can trigger the feed until the PNR is purged from the reservation system (Central System), and even if all air segments are marked as flown.

In the case of a **split**, the only way to trigger the PNR Data Feed for the parent or the child is to select the split as a triggering condition in 0 Annex 2: PNR Data elements choice and granularity.

However, if some updates are performed after the split entry, but before filing the child or end transacting the parent, those updates can also trigger the PNR Data Feed.

In addition to the triggering conditions, Amadeus can filter the sending of some PNRs. This filter is defined according to the last updatator office. In this case, updates on a PNR will trigger the feed only/except if the update is done in a given list of offices, or if the last updatator office matches a certain pattern (ex. ***6X0***). This filter should be combined with an AND to all the triggering conditions chosen in Annex 2: PNR Data elements choice and granularity⁸.

⁶ In case of a full feed (ie we send a message upon each creation/deletion/modification of elements that can trigger the feed), this is not very interesting. However, if a client wants to receive the feed for example only if an SSR FQTV is present in the PNR, then we would combine the triggering rules with the presence of an SSR FQTV.

⁷ Element value is SSR code for SSRs, SK code for keywords, a given string of characters matching the beginning of the freetext for OSIs, remark category for RM remarks.

⁸ The receiver must define in its implementation document the nature of the triggering condition (only or except) as well as the list of allowed offices (in the case of a Only condition) or forbidden offices (in the case of a Except condition).

As a consequence, with this filter, Amadeus can avoid sending the updates made by the client itself.

Examples

Example 19– Trigger due to an action at element level (Addition)

Assumption: 6X is receiving a PNR Data Feed where eligibility is: PNRs containing a 6X segment.

Triggering conditions are all the addition/modification/deletion on any elements parts of Annex 2: PNR Data elements choice and granularity.

Original PNR. This PNR is not eligible for 6X at the moment.

```

--- RLR ---
RP/MUC1A0701/MUC1A0701          LS/SU   1AUG05/1431Z   X8M9VA
1. TEST/A
2  7X 421 J 21SEP 3 SYDMEL HK1      0930 1055   *1A/E*
3  AP 0123456789-H
4  TK OK01AUG/MUC1A0701
  
```

Same PNR after modifications. A 6X Segment was added. This PNR will be part of the 6X feed because a 6X segment was added (Triggering element: Itinerary, Action: Addition).

```

--- RLR ---
RP/MUC1A0701/MUC1A0701          LS/SU   1AUG05/1433Z   X8M9VA
1. TEST/A
2  6X 016 F 20SEP 2 SYDLHR HK1      1  1620 0455+1 *1A/
3  7X 421 J 21SEP 3 SYDMEL HK1      0930 1055   *1A/E*
4  AP 0123456789-H
5  TK OK01AUG/MUC1A0701
  
```

Example 20 – Trigger due to an action at element level (Deletion)

Assumption: 6X is receiving a PNR Data Feed where eligibility is: PNRs containing a 6X segment.

Triggering conditions are all the addition/modification/deletion on any elements parts of Annex 2: PNR Data elements choice and granularity.

The previous PNR is modified again and the 6X Segment is deleted. The PNR becomes no more eligible for 6X, but it will trigger 6X feed one more time as a triggering element was deleted (Trigger Element: Itinerary, Action: Deletion).

```

--- RLR ---
RP/MUC1A0701/MUC1A0701          LS/SU   1AUG05/1436Z   X8M9VA
1. TEST/A
2  7X 421 J 21SEP 3 SYDMEL HK1      0930 1055   *1A/E*
3  AP 0123456789-H
4  TK OK01AUG/MUC1A0701
  
```

Example 21 – No trigger due to flown segments

Assumption: 6X and 7X are receiving a PNR Data Feed where eligibility is respectively: PNRs containing a 6X segment including elements associated to flown segments, PNR containing a 7X segment including elements associated to flown segments.

Triggering conditions are all the addition/modification/deletion on any elements parts of Annex 2: PNR Data elements choice and granularity for 6X and 7X.

Original PNR:

RP/MUC1A0701/MUC1A0701	AF/RM	27SEP05/1249Z	YE9RDM
1. TEST/MR			
2 7X6220 Y 25SEP 2 ORYNCE	FLWN		
3 6X 009 Y 27SEP 2 LHRSYD HK1	4	2120 0610+2	*1A/
4 AP MUC - AMADEUS DEFAULT OFFICE - A			
5 TK OK20SEP/MUC1A0701			

A new contact information is added. This will not trigger a 7X feed because all 7X segments are flown. However it triggers a 6X feed because segment 3 is not flown yet.

RP/MUC1A0701/MUC1A0701	AF/RM	27SEP05/1249Z	YE9RDM
1. TEST/MR			
2 7X6220 Y 25SEP 2 ORYNCE	FLWN		
3 6X 009 Y 27SEP 2 LHRSYD HK1	4	2120 0610+2	*1A/
4 AP MUC - AMADEUS DEFAULT OFFICE - A			
5 AP 0123456789-H			
6 TK OK20SEP/MUC1A0701			

Example 22 – Split

Assumption: 6X is receiving a PNR Data Feed where eligibility is: PNRs containing a 6X segment.

Case 1: Triggering rules is any modification on a segment.

Case2: Triggering rules is any modification on a segment OR split.

Original PNR:

RP/MUC1A0701/MUC1A0701	AF/RM	27SEP05/1249Z	YE9RDM
1. TEST/MR 2. TEST/MRS			
3 6X 009 Y 27SEP 2 LHRSYD HK2	4	2120 0610+2	*1A/
4 AP MUC - AMADEUS DEFAULT OFFICE - A			
5 TK OK20SEP/MUC1A0701			

If the above PNR is split, the parent and the children will be eligible for the PNR Data Feed to 6X (as they contain a 6X segment), but the parent and the children are sent only in case 2, even if 6X segments are modified (change in the number in party for the parent, and creation for the children).

Example 23 – Trigger due to an action at element value level

Assumption: 6X is receiving a PNR Data Feed where eligibility is: PNRs containing a 6X segment.

Triggering rules is: Addition of SSR VGML (element: SSR, value: VGML)

Original PNR. This PNR is eligible for 6X as it contains a 6X segment.

--- RLR ---			
RP/MUC1A0701/MUC1A0701	LS/SU	1AUG05/1441Z	X8M9VA

```
1. TEST/A
2  7X 421 J 21SEP 3 SYDMEL HK1      0930 1055  *1A/E*
3  6X3075 C 20DEC 2 LHRNCE HK1      1005 1305  *1A/E*
4 AP 0123456789-H
5 TK OK01AUG/MUC1A0701
```

Same PNR after modifications: a vegetarian meal was added on a 6X segment. This action will trigger a 6X feed (Trigger element: 6X associated VGML SSR, Action: Addition).

```
--- RLR ---
RP/MUC1A0701/MUC1A0701      LS/SU  1AUG05/1442Z  X8M9VA
1. TEST/A
2  7X 421 J 21SEP 3 SYDMEL HK1      0930 1055  *1A/E*
3  6X3075 C 20DEC 2 LHRNCE HK1      1005 1305  *1A/E*
4 AP 0123456789-H
5 TK OK01AUG/MUC1A0701
6 SSR VGML 6X HK1/S3
```

Example 24 – Combination of two rules (AND)

Assumption: 6X is receiving a PNR Data Feed where eligibility is: PNRs containing a 6X segment.

Triggering rules is:

Presence of a SSR FQTV

AND

Addition/modification/deletion on any elements parts of Annex 2: PNR Data elements choice and granularity.

Step 1: SSR FQTV is not present. No feed is triggered

```
RP/MUC1A0701/MUC1A0701      BM/PR  1DEC05/1326Z  2W96M2
1. TEST/A MR
2  6X6227 Y 01DEC 4 NCEORY HK1  1745 2  1805 1930  *1A/E*
3 AP MUC - AMADEUS DEFAULT OFFICE - A
4 TK OK01DEC/MUC1A0701
```

Step 2: SSR FQTV is added: The feed is triggered

```
RP/MUC1A0701/MUC1A0701      BM/PR  1DEC05/1326Z  2W96M2
1. TEST/A MR
2  6X6227 Y 01DEC 4 NCEORY HK1  1745 2  1805 1930  *1A/E*
3 AP MUC - AMADEUS DEFAULT OFFICE - A
4 TK OK01DEC/MUC1A0701
5 SSR FQTV 6X HK/ 6X123456789/6
```

Step 3: SSR FQTV is present and a remark is added. The feed is triggered.

```
RP/MUC1A0701/MUC1A0701      BM/PR  1DEC05/1333Z  2W96M2
1. TEST/A MR
2  6X6227 Y 01DEC 4 NCEORY HK1  1745 2  1805 1930  *1A/E*
3 AP MUC - AMADEUS DEFAULT OFFICE - A
4 TK OK01DEC/MUC1A0701
5 SSR FQTV 6X HK/ 6X123456789/6
6 RM THIS IS A REMARK
```

Example 25 – Last Updator office filter

Triggering rules is:

Forbidden Last updator office is NCE6X0980

AND

Addition/modification/deletion on any elements parts of Annex 2: PNR Data elements choice and granularity.

Step 1: We are in MUC1A0701. The feed is triggered

RP/MUC1A0701/MUC1A0701	BM/PR	1DEC05/1326Z	2W96M2
1. TEST/A MR			
2 6X6227 Y 01DEC 4 NCEORY HK1 1745 2 1805 1930 *1A/E*			
3 AP MUC - AMADEUS DEFAULT OFFICE - A			
4 TK OK01DEC/MUC1A0701			

Addition of a remark: feed is triggered

RP/MUC1A0701/MUC1A0701	BM/PR	1DEC05/1326Z	2W96M2
1. TEST/A MR			
2 6X6227 Y 01DEC 4 NCEORY HK1 1745 2 1805 1930 *1A/E*			
3 AP MUC - AMADEUS DEFAULT OFFICE - A			
4 TK OK01DEC/MUC1A0701			
5 RM TEST			

Step 2: The same PNR is retrieved in NCE6X0980.

RP/MUC1A0701/MUC1A0701	BM/PR	1DEC05/1326Z	2W96M2
1. TEST/A MR			
2 6X6227 Y 01DEC 4 NCEORY HK1 1745 2 1805 1930 *1A/E*			
3 AP MUC - AMADEUS DEFAULT OFFICE - A			
4 TK OK01DEC/MUC1A0701			
5 RM TEST			

Modification of the remark: The feed is not triggered.

RP/MUC1A0701/MUC1A0701	BM/PR	1DEC05/1326Z	2W96M2
1. TEST/A MR			
2 6X6227 Y 01DEC 4 NCEORY HK1 1745 2 1805 1930 *1A/E*			
3 AP MUC - AMADEUS DEFAULT OFFICE - A			
4 TK OK01DEC/MUC1A0701			
5 RM TEST DU LAST UPDATOR FILTER			

Step 3: The same PNR is retrieved in NCE6X0981.

RP/MUC1A0701/MUC1A0701	BM/PR	1DEC05/1326Z	2W96M2
1. TEST/A MR			
2 6X6227 Y 01DEC 4 NCEORY HK1 1745 2 1805 1930 *1A/E*			
3 AP MUC - AMADEUS DEFAULT OFFICE - A			
4 TK OK01DEC/MUC1A0701			
5 RM TEST DU LAST UPDATOR FILTER			

Modification of the remark: The feed is triggered.

RP/MUC1A0701/MUC1A0701	BM/PR	1DEC05/1326Z	2W96M2
------------------------	-------	--------------	--------

1. TEST/A MR
2 6X6227 Y 01DEC 4 NCEORY HK1 1745 2 1805 1930 *1A/E*
3 AP MUC - AMADEUS DEFAULT OFFICE - A
4 TK OK01DEC/MUC1A0701
5 RM TEST DU LAST UPDATOR FILTER ON ALLOWED OFFICE

Output Selection

Definition

If a PNR has verified at least one triggering condition, its view will be sent to the client via the PNR Data Feed. However, clients may not be interested in all the elements available in the PNR view.

Then the view can also be restricted by clients to a set of elements to be chosen in 0 Annex 2: PNR Data elements choice and granularity

The elements of the output selection can be different from the trigger data elements selection.

Credit Card Concealment

Several dedicated elements in the SBR can contain credit card numbers (FP element, SSR FOID, SSR EPAY, Miscellaneous documents form of payment, Automatic Hotel segment, Automatic Car segmen). The client can choose to receive credit cards located in these elements concealed or unconcealed. Credit card numbers located in remarks and or other not relevant elements are not considered.

By default the PNR Data Feed is set up to send credit card number concealed.

Activity

Warning: Activity is not available for message versions under 7.1.

Pre-condition: Activity is only available for interactive feed (online or deferred).

The message sent through the feed can contain, in addition of the image of the PNR sent, an indication of the activity on the elements⁹ of the PNR since the last EOT. (This is not the history of the PNR. History is not sent through the PNR Data Feed).

For each element¹⁰ modified since last EOT, the message should contain a reference to this element and a flag to indicate if the element has been added, deleted¹¹ or modified. In case of modification or deletion of an element, the message also contains the old value of this element (ie the full element as it was represented in the envelope before its modification/deletion).

NB: In the case of a split, the only activity available is “split” for the parent and for the child.

⁹ Activity is available only for elements selected to be part of the output in Annex 2: PNR Data elements choice and granularity

¹⁰ Activity is available only for elements selected to be part of the output in Annex 2: PNR Data elements choice and granularity

¹¹ Deletion means that the element has been transferred in the history of the PNR, and is no more part of the face of the PNR. It is not linked to active or inactive status codes (For segments or SSRs for example)

However, if some updates are performed after the split entry, but before filing the child or end transacting the parent, those updates will also appear in the activity.

NB: In case of PNR replication, a “replication” process activity is reported for the newly created PNR. Moreover if the replication entry used is RRA, an activity is also reported in the original PNR.

NB: For Miscellaneous Documents, the old value is not available in case of modification or deletion.

NB: The only activity reported in case of change in the header of the PNR is a change in the RP line (owner office field or queuing office field)

NB: In the case of an update in a PNR by a reaccommodation process (TTY or Altea Inventory), a reaccommodation indicator is added in the SBRRES message.

NB: In an OA PNR, if an air segment is updated with an inactive status, it is directly placed in the history of the PNR upon a simple EOT (where an ETK or ERK is needed on a 1A PNR). So the segment is considered as deleted. However the old value of the segment sent through the PNR Data Feed will not be the value of the segment at the previous envelope, but the segment with its inactive status code. This exception allows clients to receive this inactive status code through the PNR Data Feed. The same behavior applies also for SSRs with a status code.

NB: In case of an automatic segment cancellation due to the MFQ-ETTL process, the PNR Data Feed will report this activity as a process activity.

NB: In some exceptional cases the PNR itself may be corrupted.-In this scenario, as the customer may have missed some SBRRES (corresponding to all the envelopes of the PNR during its corruption), the best solution is to provide the latest image of the PNR in a particular mode. In that case, whatever the actions made on the PNR, the only activity reported is a special REC process activity.

By default this activity/trigger will be set to "do not receive".
However, if there is some value for the customer to receive corrupt PNRs, at implementation time, the customer will be able to choose to receive these specific SBRRES messages.

Example 26 – Examples for the activity

Assumption: 6X is receiving a PNR Data Feed where eligibility is: PNRs containing a 6X segment.

Triggering rules are any addition/modification/deletion on any elements parts of Annex 2: PNR Data elements choice and granularity.

Output is all the elements part of Annex 2: PNR Data elements choice and granularity.

Step 1: Creation of a PNR

```

--- RLR ---
RP/NCE6X0100/NCE6X0100          AA/GS   6APR06/1210Z   ZED5WI
1. TEST/A
2  6X 798 Y 06APR 4 LHRHEL HK1 1605 1930 06APR E 6X/
3 AP NCE 111222333 - ABG PMG DEMO OFF NCE - A
4 TK OK06APR/NCE6X0100
5 RM THIS IS A REMARK

```

For step 1, the current content of the PNR is sent through the feed.
The following activity is also reported:

- Addition of a name + reference to the name
- Addition of a segment + reference to the segment
- Addition of a contact element + reference to the element
- Addition of a TK element + reference to the element
- Addition of a remark element + reference to the element

Step 2: Remark is changed, previous segment is deleted, an other segment is added

```

--- RLR ---
RP/NCE6X0100/NCE6X0100          AA/GS   6APR06/1218Z   ZED5WI
1. TEST/A
2  6X 698 Y 06APR 4 LHRJFK HK1  1235 1515  06APR      6X/
3 AP NCE 111222333 - ABG PMG DEMO OFF NCE - A
4 TK OK06APR/NCE6X0100
5 RM REMARK CHANGED
  
```

For step 2, the current content of the PNR is sent through the feed.
The following activity is also reported:

- Deletion of a segment + reference of the old segment + old segment
- Addition of a segment + reference of the new segment
- Change in the remark element + reference to the remark element + old remark

Example 27 – Examples for the activity and split

Assumption: 6X is receiving a PNR Data Feed where eligibility is: PNRs containing a 6X segment.

Triggering rules are any addition/modification/deletion on any elements parts of Annex 2: PNR Data elements choice and granularity.

Output is all the elements part of Annex 2: PNR Data elements choice and granularity.

Step 1: Creation of a PNR

```

--- RLR ---
RP/NCE6X0100/NCE6X0100          AA/GS   6APR06/1231Z   ZED6LY
1. TEST/A  2. TEST/B
3  6X 798 Y 06APR 4 LHRHEL HK2  1605 1930  06APR   E  6X/
4 AP NCE 111222333 - ABG PMG DEMO OFF NCE - A
5 TK OK06APR/NCE6X0100
  
```

For step 1, the current content of the PNR is sent through the PNR Data Feed.
The following activity is also reported:

- Addition of a name + reference to the first name
- Addition of a name + reference to the second name
- Addition of a segment + reference to the segment
- Addition of a contact element + reference to the element
- Addition of a TK element + reference to the element

Step 2: Split + updates during the split

After split entry, but before filing the child PNR a remark is added in the child PNR. Then the parent PNR is end transacted.

```

--- AXR RLR ---
RP/NCE6X0100/NCE6X0100          AA/GS   6APR06/1233Z   ZED6LY
1. TEST/B
  
```

```

2  6X 798 Y 06APR 4 LHRHEL HK1 1605 1930 06APR E 6X/
3  AP NCE 111222333 - ABG PMG DEMO OFF NCE - A
4  TK 0K06APR/NCE6X0100
*  SP 06APR/AAGS/NCE6X0100-ZED6PK

```

The current face of the parent PNR is sent through the PNR Data Feed
The only activity reported is: PNR has been split.

```

--- AXR RLR ---
RP/NCE6X0100/NCE6X0100          AA/GS  6APR06/1233Z  ZED6PK
1. TEST/A
2  6X 798 Y 06APR 4 LHRHEL HK1 1605 1930 06APR E 6X/
3  AP NCE 111222333 - ABG PMG DEMO OFF NCE - A
4  TK 0K06APR/NCE6X0100
5  RM THIS IS A REMARK
*  SP 06APR/AAGS/NCE6X0100-ZED6LY

```

The current face of the Child PNR is sent through the PNR Data Feed.
The activity reported is:
 PNR has been split
 Addition of a remark + reference to the element

Example 28 – Examples for the activity and inactive status code

Assumption: 6X is receiving a PNR Data Feed where eligibility is: PNRs containing a 6X segment.
Triggering rules are any addition/modification/deletion on any elements parts of Annex 2: PNR Data elements choice and granularity.
Output is all the elements part of Annex 2: PNR Data elements choice and granularity.

Case 1: 1A PNR

Original PNR:

```

--- RLR ---
RP/MUC1A0701/MUC1A0701          BM/PR  2AUG06/1338Z  2MKT63
1. DURAND/PAUL
2  6X1442 Y 20SEP 3 LHREDI HK1 1115 1235 20SEP E
3  AP MUC - AMADEUS DEFAULT OFFICE - A
4  TK 0K02AUG/MUC1A0701

```

PNR updated:

```

--- RLR ---
RP/MUC1A0701/MUC1A0701          BA/RM  2AUG06/1342Z  2MKT63
1. DURAND/PAUL
2  6X1442 Y 20SEP 3 LHREDI UN1 1115 1235 20SEP E
3  6X1446 Y 20SEP 3 LHREDI TK1 1315 1435 20SEP E
4  AP MUC - AMADEUS DEFAULT OFFICE - A
5  TK 0K02AUG/MUC1A0701

```

A reaccommodation is performed because 6X1442 is no more operational. The passenger is placed on 6X1446.
Hence the following activity is reported:

Process activity: Reaccommodation

Modification of a segment + reference to the segment 6X1442 (old value = 6X1442 Y 20SEP 3 LHREDI **HK1** 1115 1235 20SEP)

Addition of a segment + reference to the segment 6X1446

Case 2: OA PNR (inactive segment)

Original PNR:

```

--- RLR ---
RP/SWI1G/SWI1GVER16                2AUG06/1340Z    2MKUJW
1. DUPONT/TOM
2  6X1442 Y 20SEP 3 LHREDI HK1  1115 1235  20SEP  E
  
```

PNR updated:

```

--- RLR ---
RP/SWI1G/SWI1GVER16                2AUG06/1342Z    2MKUJW
1. DUPONT/TOM
2  6X1446 Y 20SEP 3 LHREDI HK1  1315 1435  20SEP  E
  
```

A reaccommodation is performed because 6X1442 is no more operational. The passenger is placed on 6X1446.

In the history of the PNR, we can see:

```

001/002 CS/6X1442 Y 20SEP 3 LHREDI UN1 1115 1235/HK    E*
      002 AS/6X1446 Y 20SEP 3 LHREDI TK1 1315 1435/TK    E*
  
```

Hence the following activity is reported:

Process activity: Reaccommodation

Deletion of a segment + reference to the deleted segment 6X1442 (old value = BA1442 Y 20SEP 3 LHREDI **UN1** 1115 1235 20SEP)

Addition of a segment + reference to the segment 6X1446

Case 3: 1A PNR (inactive SSR)

Original PNR:

```

--- RLR ---
RP/MUC1A0701/MUC1A0701            BM/PR    2AUG06/1338Z    2MKT63
1. DURAND/PAUL
2  6X1442 Y 20SEP 3 LHREDI HK1  1115 1235  20SEP  E
3 AP MUC - AMADEUS DEFAULT OFFICE - A
4 TK OK02AUG/MUC1A0701
5 SSR WCHR 6X HN1
  
```

PNR updated with a SSR status change:

```

--- RLR ---
RP/MUC1A0701/MUC1A0701            BM/PR    2AUG06/1338Z    2MKT63
1. DURAND/PAUL
2  6X1442 Y 20SEP 3 LHREDI HK1  1115 1235  20SEP  E
3 AP MUC - AMADEUS DEFAULT OFFICE - A
4 TK OK02AUG/MUC1A0701
5 SSR WCHR 6X UN1
  
```

Hence the following activity is reported:

Modification of an element + reference to the SSR WCHR (old value = SSR WCHR 6X **HN1**)

Case 4: OA PNR (inactive SSR)

Original PNR:

```

--- RLR ---
RP/SWI1G/SWI1GVER16                2AUG06/1340Z    2MKUJW
1.DUPONT/TOM
2  6X1442 Y 20SEP 3 LHREDI HK1  1115 1235  20SEP  E
3  SSR WCHR 6X HN1

```

PNR updated with a SSR status change:

```

--- RLR ---
RP/SWI1G/SWI1GVER16                2AUG06/1342Z    2MKUJW
1.DUPONT/TOM
2  6X1446 Y 20SEP 3 LHREDI HK1  1315 1435  20SEP  E

```

Status of the SSR is changed to UN and PNR is end transacted

In the history of the PNR, we can see:

```

009/012 SX/SSR WCHRBAUN1/HN/6X1442 Y 20SEP LHREDI

```

Hence the following activity is reported:

Deletion of an element + reference to the deleted SSR WCHR (old value = SSR WCHR 6X **UN1**)

Feed type

Online interactive feed (Online feed)

The interactive feed mechanism sends a PNR image immediately after the End of Transaction of the PNR if it is eligible and if a triggering condition is verified.

Batch Feed Last image deferred feed (Classical batch feed)

If configured in this mode the PNR Data feed sends only the last image of a PNR (even if this one has been updated several times since the last run).

The last image deferred feed stores the unique reference of the PNR after each EOT. On a regular basis, the process uses this list of reference in order to send all the PNRs view listed. The PNR image is retrieved at batch execution time, nevertheless the "to be sent" PNR list is built on a continuous basis (each relevant EOT on the PNR puts the PNR in the list).

For instance, if a PNR has been updated 3 times since the last run, only the last PNR image will be sent.

Example 29 – Feed type: Only last image since last feed

Same original PNR as for the previous example, followed by the same 3 modifications. The only PNR image that will be sent will be the last one:

```

--- RLR ---
RP/MUC1A0701/MUC1A0701            LS/SU  1AUG05/1537Z    X8M9VA
1.TEST/A
2  7X 421 J 21SEP 3 SYDMEL HK1      0930 1055  *1A/E*

```

3 AP NEW CONTACT ELEMENT-H

4 TK OK01AUG/MUC1A0701

Media

The media corresponds to the technical characteristics of the data transfer solution. Two different media are available for the PNR Data feed: FTP and MQ.

FTP

The File Transfer Protocol is the solution mainly used to transmit the data for the deferred batch feed.

The whole file containing all PNRs is sent at the pre-defined time.

In case the data transfer fails, the data will be re-sent later when the communication issue is tackled.

In case the generation of the file is delayed (by internal or external issues) the system must be able to deliver the data later on.

MQ series

The MQ series solution is used to transmit messages containing each a PNR image to be sent. It is used within the context of the online PNR Data Feed.

It is a guaranteed delivery mechanism provided by IBM.

The customer has to pick up the message in the queue where the PNR Data Feed put the message.

Log

The PNR record locator, creation date and envelope number/send date of the PNR images sent have to be logged in order to allow retrieving which PNR images have been sent to each customer, and this for a one year period.

This can be used for audit and billing purpose

Recovery procedures

This paragraph describes the procedures that Amadeus can execute in case of failure/corruption linked to the PNR Data Feed.

Problems may occur on the link between Amadeus and the client, on client's side or on Amadeus' side.

Link problem

FTP

If the FTP file transfer fails, the file will be resent later when the link is up again.

MQ

The PNR data feed uses a guaranteed delivery mechanism through a MQ link. So if the messages are not sent due to a link problem, they will be queued and sent when the link is up again.

Failure on client's side

This is the case where Amadeus has sent the messages successfully through the feed, but the messages are lost or corrupted on client's side.

NB: Detailed procedures describing how to request the activation of the first or second level recovery have to be written in the Implementation document associated to each PNR Data Feed.

To restart with a clean database, Amadeus can launch a new initialization procedure. The format (EDIFACT/XML) and the version of the messages used for the initialization file will be the same as the one used for the regular PNR Data Feed. The file will be sent via FTP (or via any other media agreed with the client (CD, DVD for example)).

However if the client needs to receive images of the PNRs lost during the failure, Amadeus can provide the client with the first level recovery files, depending on the failure duration.

First Level Recovery

Amadeus keeps a copy of all the messages sent through the PNR Data Feed during three days by default¹².

Hence Amadeus can send again to the client all the messages that have been sent through their PNR Data Feed publication during a given period between the day of the first level recovery execution and three days before.

The format used (XML or EDIFACT) and the SBRRES version used will be the one used usually in the PNR Data Feed (as we just send again the original messages we have already sent)

To build the first level recovery file, the client needs to provide us with:

- Begin Date/time

- End Date/time

- Feed name (if the client receives several feeds)

Online feed

In the case of an online feed, all the PNRs to resend will be grouped in one file (same format as for a normal batch file) that will be sent manually via FTP (or via any other media agreed with the client (CD, DVD for example)).

Alternatively, the PNRs to resend can be resent via MQ (same format as a normal online feed). But in this case, there won't be any way to differentiate the normal traffic from the recovery traffic!

Batch feed

We can send again the files that have already been built.

Example 30 – First level recovery

Client 6X has a corruption on its side between 19MAR and 20MAR.

Case 1: On 20MAR, 6X opens a Work Order (WO) asking 1A to send again the SBRRES sent between 19MAR and 20MAR. WO is taken into account on 21MAR, which is 3 days after the beginning of the corruption period. The first level recovery procedure can be successfully executed.

¹² The period for which the data sent to the customer is kept on Amadeus depends on each PNR Data feed implementation as it is driven by customer need, and as the cost per retention period unit is dependent on the volume of data sent.

Case 2: On 21MAR, 6X opens a WO asking 1A to send again the SBRRES sent between 19MAR and 20MAR. WO is taken into account on 22MAR, which is 4 days after the beginning of the corruption. It is too late for first level recovery!

Second level recovery batch feed

Second level recovery batch feed should be used in case of failures longer than what is supported by the first level recovery for an interactive online feed.

Before the failure, a batch feed (last image differed feed) has to be set up on the production environment. This batch feed is ready to be activated in case of failure.

In case of failure, if the client estimates that the failure will last more than the first level recovery period, Amadeus can activate the second level recovery batch feed.

Each day at a fixed time when the second level recovery batch feed is activated, 1A will generate a file containing the last image of the active PNRs that would have been sent through the regular online feed during the day.

Note that the PNRs that have been purged when the file is built cannot be included in the file. Hence the file has to be generated just before the daily purge process to minimize the number of PNRs lost.

The format of the file is be the same as the one used for initialization file or classic batch files, and described in the SBRRES mapping document. The version of the SBRRES message for the recovery batch feed will be the same as the version used in the regular feed.

If necessary the resulting file can be split into several smaller files.

Once generated, the file(s) will be available on the Amadeus FTS server.

Example 31 – Second level recovery batch feed

Please note that the times given in those examples are just indicative. To avoid loosing any PNR, the client must ensure to warn 1A of their failure with enough notice.

Note: the following example focuses on the second level recovery batch feed, because first level and second level recovery are independent process.

20JUN at 08:00GMT: A failure occurs on 6X's side, without affecting 1A (SBRRES messages are still sent successfully to 6X).

21JUN at 09:00GMT: 6X detects the failure.

1A does not interrupt the PNR Data Feed to 6X because parts of 6X robotics can still receive and process SBRRES messages.

21JUN at 12:00GMT: 6X informs 1A that the failure will not be solved shortly and request the activation of the second level recovery batch feed.

21JUN at 18:00GMT: 1A activates the second level recovery batch feed.

21JUN at 20:00GMT: 1A builds the second level recovery batch file for the period from 21JUN at 18:00 to 21JUN at 20:00GMT. Once generated, the file is available for 6X on the FTS server.

22JUN at 20:00GMT: 1A builds the second level recovery batch file for the period from 21JUN at 20:00 to 22JUN at 20:00GMT. Once generated, the file is available for 6X on the FTS server.

23JUN at 10:00GMT: 6X informs 1A that the failure is now solved.

23JUN at 20:00GMT: 1A builds the last second level recovery batch file for the period from 22JUN at 20:00 to 23JUN at 10:00GMT. Once generated, the file is available for 6X on the FTS server.

Failure on Amadeus' side

A new initialization procedure must be done because in this case we may not have the log file containing the references to the PNRs that should have been sent through the PNR Data Feed. The initialization file will be available in XML or EDIFACT format and will be sent via FTP (or via any other media agreed with the client (CD, DVD for example)).

Data format

The data format corresponds to the way the PNR information is encoded. The EDIFACT industry standard is the main one.

EDIFACT

The EDIFACT industry standard could be used to convey PNR information in a structured way to the customer.

The message used by the PNR Data Feed is the SBRRES.

XML

The information could as well be formatted in XML. However, the volume of the information in compressed XML is about 20 times the volume of the same information formatted in EDIFACT.

Examples and Scenarios

Classical full airline feed

Parameters

Client: 6X

Type of feed: Interactive online feed

Eligibility rules:

PNR contains a 6X segment

Triggering rules:

Any addition/modification/deletion of any element

Scenario

<i>Envelope</i>	<i>Action</i>	<i>Triggers the feed</i>
0	PNR creation: the PNR contains a 7X segment but no 6X segment	No
1	Addition of a remark (RM)	No
2	A 6X segment is added	Yes
3	Addition of a SSR associated to the 7X segment	No
4	Addition of an SSR associated to the 6X segment	Yes
5	Addition of an OSI YY	Yes
6	Addition of a remark (RM)	Yes
7	Deletion of the 7X segment	Yes
8	Addition of a 8X segment	Yes
9	Deletion of the 6X segment	Yes

10 Any modification (except a 6X segment addition) No
To sum up, as the feed is an interactive online feed, 6X will receive the PNR images for envelope 2, 4, 5, 6, 7, 8 and 9

Office feed

Parameters

Client: 7X
Type of feed: Last image deferred feed.
Frequency: Every day at 06AM GMT
Eligibility rules:
 Owner of the PNR is a 7X ATO/CTO (***7X0***)
Triggering rules:
 Any addition/modification/deletion of itinerary, SSR

Scenario

<i>Envelope / Time</i>	<i>Action</i>	<i>Triggers the feed</i>
0-20FEB07AM	PNR creation: PNR created in office NCE7X0100, containing a 6X segment	Yes
1-20FEB08AM	Addition of a remark (RM)	Yes
2-20FEB07PM	Addition of an OSI 6X	No
3-21FEB10AM	Addition of an SSR VGML 6X	Yes
4-21FEB08PM	Addition of a 7X segment	Yes
5-22FEB09AM	Addition of an 8X segment	Yes
6-22FEB10AM	Deletion of the 8X segment	Yes
7-23FEB11AM	Adding a 6X segment and changing owner office to NCE6X0200	Yes
8-23FEB02PM	Any modification (except changing the owner office back to 7X)	No

To sum up:
On 21FEB-06AM, 7X receives envelope 2
On 22FEB-06AM, 7X receives envelope 4
On 23FEB-06AM, 7X receives envelope 6. Note that 7X won't know that the 8X segment has been added then deleted.
On 24FEB-06AM, 7X will receive envelope 8. Note that the PNR is no more eligible then and won't be sent anymore to 7X if the owner office is not changed back to 7X.

Light airline feed

Parameters

Client: 6X
Type of feed: Interactive online feed
Eligibility rules:
 PNR contains a 6X segment

Triggering rules:

Any addition/modification/deletion of itinerary, SSR, OSI

Scenario

<i>Envelope</i>	<i>Action</i>	<i>Triggers the feed</i>
0	PNR creation: the PNR contains a 7X segment but no 6X segment	No
1	Addition of an OSI 7X	No
2	A 6X segment is added	Yes
3	Addition of a SSR associated to the 7X segment	No
4	Addition of a SSR associated to the 6X segment	Yes
5	Addition of an OSI YY	Yes
6	Addition of a remark (RM) (non associated, or associated to a 6X segment)	No
7	Addition of a contact element (AP)	No
8	Addition of a 8X segment	Yes
9	Deletion of the 6X segment	Yes
10	Any modification (except a 6X segment addition)	No

To sum up, as the feed is an interactive online feed, 6X will receive the PNR images for envelope 2, 4, 5, 8 and 9

Annex 1: PNR data elements description

This annex describes the functional PNR data elements that can be included in the PNR Data Feed **when they are available**. The PNR data elements are used in this product in order to assess the PNR selection, trigger the inclusion of a PNR view in the feed, to be sent to the client.

PNR Header

This item is mandatory and is present for every PNR sent.
The following information is part of the PNR Header

- **PNR record locator (Amadeus reservation control number)**
- **OA record Locator (Other GDS reservation control number)**
- **Booking Originator record Locator and Point of Sell**
- **Envelope number**
- **PNR Creation Date**
- **PNR Last update Date/Time**
- **PNR Purge Date**
- **PNR type (OA, Group, Non commercial, Corporate...)**
- **Marketing PNR references (In case of operating PNR)**
- **Queuing office at feed time**
This is the queuing office at PNR Data Feed time; it could be either an Amadeus Office (ex: NCE1A0950) or an OA office (ex: HDQ1S).
- **Number of passengers in the PNR**
- **Point of Sale of the owner of the PNR (called Point of Sale of the PNR), PoS of the Last Updator of the PNR and PoS of the Creator of the PNR in Amadeus.**

Warning: PoS for the owner is not available for message versions under 6.1. PoS for the. If mandatory elements are not provided, there will be no Pos Information in message version 6.2. Last Updator and the Creator are not available for message versions under 7.1

IATA code, provided that it exists

OfficeID (May be a pseudo office ID in case of 3rd party booking). If not provided, default value.

Agent details (numeric sine, initials, duty code), provided that it exists

Originator (A/Airline, T/Travel Agent, E/Electronic selling system, N/Non IATA,...). provided that it exists.

System used (1A, 1G, 1S) - if not provided, default value.

True City (3 letters) provided that it exists,

Country (2 letters) - if not provided, default value.

Passenger information

This is available for all the passengers present in the PNR.

NM/NG – Name/Group Name element¹³

- *Last name*
- *First name and title*
- *Extra-seat indicator*
- *Cabin baggage indicator*
- *Staff type*
- *Infant first name and last name*
- *Passenger type codes*

Additionally for Group PNR:

- *Group name*
- *Number of unassigned passenger (within a group)*

Itinerary

Air Itinerary

Different types of air segments can be sent: Normal (including Passive and Ghosts), Open, Informational, ARNK.

- **Segment creation date and time**
- **Level of access used for sell**
Amadeus Access Sell / Direct Access on carrier / Allotment
- **Segment type**
Marketing/Operating
- **Flight number**

It is made of the airline code and the flight number. In case the flight has an alphabetical suffix, the suffix is included as well. In case, of joint flight number, the code of the second airline is included.

For an open segment, the flight number is replaced by "OPEN".

- **Reservation status**

It is the status code of the segment (Ex: HK, HN, HL...).

This is not applicable in case of open segment.

- **Flown status indicator**

In case the flight is flown, a flown indicator is sent.

An active segment becomes flown 2 days after its departure date.

Not applicable to open segment.

- **Class of service**

¹³ NZ (non commercial PNRs) and NC (corporate PNRs) are handled as group PNRs in the PNR data feed.

This is the booking class; if relevant the night indicator is transmitted as well when it is a night class.

- **Number of Passengers**

This is the number in party for the segment.
It is not applicable to open segment.

- **Day of operation**

Day of the week (1= Monday, etc...)

- **Flight Departure Date and Time**

This is the departure date and time (hhmm) of the segment. The latest check-in time (hhmm) is provided as well (if present).

For information segment only the departure date is present.

For open segment only the departure date could be present (it is optional).

The date/time are the one as they appear in the PNRs (not the effective date/time of departure/arrival)

- **Check-in time**

- **Board Point**

Location code is provided (ex: LHR, CDG...); departure terminal identification is included (if present).

For open and information segment, there is no departure terminal identification provided.

- **Flight Arrival Date and Time**

This is the arrival date and time (hhmm) of the segment. If present the day change indicator is provided as well (values from +2 to -1). The arrival date is in sync with the departure date and the day change indicator.

It could happen that this date and time is not consistent with the schedule.

This is not applicable for information and open segments.

- **Off Point**

The arrival location code is sent (ex: JFK). Arrival terminal identification is included (if present).

- **Electronic Ticketing**

The electronic ticketing eligibility of the segment is sent.

This is not applicable to open and informational segments.

- **Inventory Data**

Inventory Point of Sale classification, CRS, country code, cabin code, sub-class number, flight type

- **CodeShare Data**

Carrier code, flight number and suffix, Codeshare agreement, class of service, flight type, Cascading indicator.

- **Yield Data**

Bid price amount, yield amount, Revenue Loss, Adjusted Yield, Effective Yield, O+D Yield, Class/class combination (of the yield)

- **Overbooking/ Oversale Indicator + overbooking reason**

- **External system record locator**

This is the reference of the segment in the airline's system (in case record locator return is supported by the airline).

This is not applicable to open and informational segments.

- **Negotiated Space Identifier**

Warning: DBID is not available for message versions under 7.1.

In case of a segment sold from a Negotiated Space Block, this is the identifier of the Block (DBID). DBID is only available for Amadeus Negotiated Space participants who are also Amadeus Altea Inventory users.

- **Marriage and connections**
Contains references to the connected and married segments (and domination indicator for married segments)
- **Point of Sale (PoS)**

Warning: PoS at segment level is not available for message versions under 6.2a.

The point of sale at segment level including (when available):

- ⇒ IATA code (Provided for 1A and non 1A bookings provided that it exists)
- ⇒ OfficeID (Provided when 1A booking, may be a pseudo office ID in case of 3rd party booking)
- ⇒ Originator (A/Airline, T/Travel Agent, E/Electronic selling system, N/Non IATA,...)
- ⇒ System used (1A, 1G, 1S,...)
- ⇒ True City code where segment was sold (3 letters)
- ⇒ Country code where the segment was sold (2 letters)

- **Operating record locator (if the segment is operated by an other airline)**
- **Original Action Code (action code used at sell time)**

Warning: Original Action Code is not available for message versions under 7.1.

- **Aircraft type (equipment code: 4 an. characters)**

Automated Car segment

Warning: Car Segments are not available for message versions under 7.3.

- **Segment type**

CCR for Car

- **Car Supplier Code**

The code of the car rental company : for example ZD for Budget.

- **Status Code AND Number of Cars**

This is the status code of the car segment. (Ex : 'HK', 'SS'...)

- **Pick-up date and time**
- **Drop-off date and time**

- **Airport Code/City Code**

This is the place where the car should be picked up. Ex: CDG

- **Airport/City Code Translation**

Ex: CHARLES DE GAULLE

- **Car Code**

It is the SIPP (or ACRISS) code of the vehicle which contains 4 alphanum. (Ex : 'ECAR')

- **Car Type Translation**

It gives the type of car which is contained in the SIPP code. (Ex: 'Economy')

- **Arrival time information**

It contains the arrival time. (Ex: 0900AM)

- **Return time information**

It contains the return time. (Ex: 0900AM)

- **Booking Agent information**

- **Billing Number Information**

The Billing Number option is used in connection with the Car Voucher. This option gives the end-user a means of entering information about where to send the invoice from the Car Rental Company (to the travel agency or to the client's company). (Ex: 'C12EF343GH23')

- **Billing Reference Information**

This option holds a reference, which may be needed by the Travel Agency and the corporate accounts to identify a specific booking (mission). (Ex: 'GK434882')

- **Booking Agency Field ID**

This option contains the IATA Number of the booking travel agent.

- **Corporate Discount ID**

This option contains a Corporate Discount ID. (Ex: '123456AB')

- **Confirmation Number Field ID**

This field contains the confirmation number (sent by the Car Rental Company) present in the segment.

- **Car Voucher Acknowledgement**

This field contains the indicator sent by the Car Rental Company to confirm the paper voucher or e-voucher.

- **E-Voucher Number**

This field contains the E-voucher number, generated by the Car Rental Company or Amadeus.

- **Drop Charge information**

This field contains the applicable one way rental drop charges returned by the Car Rental Company to Amadeus for Complete Access bookings.

- **Drop Off location information**

This field specifies the location at which the customer wishes to drop off the car, if different from the pick up point. It is composed by the code and the address of the Drop Off location.

(Ex: PARC08*SNCF GARE MONTPARNASSE 15)

- **Form of payment information**

This field indicates the form of payment use by the customer. (EX: Full Credit, CASH-EUR200...)

- **Frequent Traveller Number**

This field contains the identification of a traveler belonging to a Frequent Traveler Program.

- **Guarantee Information**

This field contains the method of payment that the customer will use at pick up time or the method of prepayment for leisure business. This information provides subscribers with the means to guarantee car bookings. (Ex: CHEQUE-EUR1000, CASH, ID ...)

- **Customer ID number**

This number is assigned by the subject Car Rental Company

- **Inclusive Tour Information**

The Tour number field is an ATC standard option. It contains the tour number of the tour in which the car segment is included.

- **Location Code Information**

This field provides the system with a means of storing in the PNR the Amadeus reference code for a non-airport location. This code does not appear when the pick-up location is an IATA recognized airport. (Ex: 'FRAC01')

- **Inter-city distance information**

This field contains the actual distance between pick-up and drop-off locations. (Ex: '546')

- **Name Information**

This Field provides a means of identifying which passenger in a PNR actually intends to rent the car.

- **Pickup location information**

This Field provides a mean to specify the Car Rental Company location when the AMADEUS cars location. (Ex: MIAC01)

- **Promotional Code Information**

This field contains the promotional code related to a Complete Access Plus Car Rental Company. (Ex: OFFER2000)

- **Base Rate Information**

This field contains the total amount for the length of the rental and the number of days that amount represents.

The Base Rate amount excludes taxes, surcharges for special equipment, insurance coverage and any additional charges that may apply. (Ex: -*XX*EUR1160.00 BR-2DY with XX: the Car Rental Company Code)

- **Rate Code Information**

This field identifies a rate within a given location/category/rate plan/car type. (Ex: Best)

- **Rate Guaranteed Information**

This field contains the rate guaranteed by the Car Rental Company for a car rate extracted from the AMADEUS Cars system.

It contains the following information:

- The Currency
- The Rate Amount: Rate of the booking
- The Mileage Charge: Mileage charge without unity. No maximum length is specified. Blank separator after the mileage charge
- The Free Mileage: For capped mileage, quantity of free mileage or Kilometers. M for miles and K for kilometers. For unlimited mileage, set to UNL.
- The Rate Plan: DY for Daily, WE for Week End, WY for Weekly, MY for Monthly
- Related to the Extra Day:
 - The Rate Amount
 - The Mileage Charge
 - The Free Mileage
- Related to the Extra Hour:
 - The Rate Amount
 - The Mileage Charge

- The Free Mileage

- **Rate Quoted Information**

The rate quoted option is an IATA standard much like the Rate Guaranteed field. It is designed to contain the rate quoted to the customer by an authorized agent. It contains exactly the information of the Rate Guaranteed Information.

- **Estimated Total Rate Information**

This field contains estimated total rate and the number of days and hours. It can contain a Tax or a Surcharge information as well.

- **Supplementary Information**

This field contains any special considerations that the Rental car company should be aware.

(Ex: BLACK COLOR)

- **Special Request Information**

This field indicates a particular type of service or equipment requested to the Car Rental Company. (Ex: TELEPHONE, SNOW CHAINS ...)

- **Tracking Field Information**

This field contains promotional coupon information, authorization number, vouchers or miscellaneous information requested by the car vendor. Tracking information, allowable codes and system processing are defined by each vendor and are vendor specific. (Ex: AUTH-000123456...)

- **Voucher Value Information**

This field can have the following voucher value: Full Credit (or FC), Nil Value (or NV), Group Day Apply (or GDA) or be a specific amount. (Ex: FULL CREDIT, EUR100.25, GDA...)

- **Accounting Information Free flow**

This field contains information used the agency accounting package. (Ex: 99902)

- **Rate Override Information**

This field contains a rate amount that override the rate amount booked.

- **Delivery Information**

This field contains the information from the location, where the Car Rental Company should deliver a vehicle before the start of a rental period.

- **Collection Information**

This field contains the information from the location, where the Car Rental Company should collect a vehicle after the start of a rental period.

- **Tax Information**

This field contains the information on the tax related to the segment, such as the tax amount in percent, the tax name, the tax inclusion (Included or Not Included).

- **Surcharge information**

This field contains the information on the surcharge related to the segment, such as the surcharge name, the surcharge amount with the currency or the percent value, the surcharge duration (PD = Per Day, PW = Per Week, PM = Per Month, PR = Per Rental), The surcharge maximum value.

- **Coverage information**

This field contains the information on the coverage related to the segment, such as the coverage type, the coverage type name, the coverage amount with the currency or the percent value, the coverage duration (PD = Per Day, PW = Per Week, PM =

Per Month, PR = Per Rental), The excess value with the currency, Liability label with the amount.

- **Vehicle Information**

This field contains information on the 'Car Make Model', such as the vehicle model , the group identifier, the number of doors, the maximum number of doors, the number of seats, the maximum number of seats, the number of bags, the volume of the boot.

- **Car Leisure information**

This field contains information related to the Leisure data in the case of a leisure segment, such as the leisure flag, the cancellation protection flag, the excess refund indicator , the prepaid indicator, the drive age , the email address.

- **FLWN/PAST information**

Flown/ past date segment tag when present in the PNR/history.

- **Passenger Association**

This contains the Passenger numbers associated to the segment.

Automated Hotel segments

Warning: Hotel Segments are not available for message versions under 7.3.

Automated Hotel segments can be active or ghost bookings and may include following group data

- **Segment type**
HHL for Hotel
- **Travel Product**
Date in and out, hotel chain and city codes
- **Segment Creation date and time**
- **Original reservation status**
Status code of the segment before the last End of transaction (Ex: HK, HN, GK...).
- **Message Action**
3 for Hotel
- **Current reservation status**
Status code of the current segment (Ex: HK, HN, GK...) – not the translated one coming from EoT.
- **Level of access used for sell**
Standard Access, Seamless.. and also RTSVCH source.
- **Hotel Property information**
Chain code, city code, property code, hotel name and fire safety compliancy indicator
- **Hotel Chain identification**
Travel Sector (=HTL), Amadeus company Code (1A), Chain code, chain name in 1A system, Access Level
- **Hotel check-in and check out dates**
Business Semantic CHK indicator and Local time indicator LT , Check-in and check-out dates for the hotel booking (Year AAAA, Month MM, Day DD).
- **Room Rate Details – Room information**
Room Type, Rate code, Hotel Category Code, Rate Code qualifier, Booking Code, Occupancy of the room and unit (=adult), room type override if not 1A room type.

- **Room Rate Details – Tariff details**
Rate type, Amount, Currency, Rate Plan Indicator (daily or total rate), Amount type (GUA for guaranteed, OVR rate override, QUO rate coded), Rate change indicator, Rate Category Associated charge type (Extra Adult, Crib, ...) associated charge amount, currency.
- **Room Rate Details – Rule Information**
Rule type = 18 for Deposit, 17 for Guarantee or 3 for Hold time → in that last case hold time value and hold time unit (hours) are provided.
Rule status type (= 18 for Deposit, 17 for Guarantee or 3 for Hold time) and Processing indicator (3 for Supplier)
- **Confirmation or Cancellation Number**
Company Id (1A if booking through Amadeus else Airline code for passive hotel segment), Confirmation number returned by the provider at sell or rebook time, Cancellation Number returned by provider at cancel time, Confirmation or Cancellation indicator (2 for Confirmation, CXL for Cancellation)
- **Booking Source Number**
- **Billable Info**
Billing number or Agency Accounting
Billing qualifier (901) or Agency Accounting code (902)
- **Customer Reference Info**
Corporate discount number, Hotel Guest Id number , Identifier (1 for Hotel Guest ID /CD for Corporate Discount number)
- **Frequent Traveler Info**
Company identification, Frequent traveler number
- **Guarantee or Deposit information**
Form of Payment (Credit Card, booking source, CD number, ...)
Payment type = 1 for Guarantee / 2 for Deposit
Service to be paid. Set to 3 per default for Hotel
Form of Payment Type : (CA for cash, CC for credit card, ...)
in case of FoP is credit card : Vendor Code, CC number and expiration date
Payment Type
- **Special options – hotel mkt lines, sales conditions , chain and hotel names**
 - Hotel Chain Marketing lines (MKT)
 - Sales conditions: Check in-out (CHK)/ commission (COM)/ cancellation policies (CXL)/rate description (DES)/ Meal (MEA) / Tax (TAX), Service Charge (SVC) /Number of Nights for the stay (NGT) / Pricing rate information (PRI) / Total Amount (TTL) / Stay information (STA) / Booking requirements (BOO) / Max advance booking conditions (ADV) / Extra (EXT).
 - Hotel Name NAM Chain name CNM ,
- **Miscellaneous Remark**
Type: RR Arrival Information, HBGV Hotel Booking Mean Info , HRTT Hotel rate type Info, HSI Hotel Supplementary Information, HSAV Hotel Non Standard Saving Amount, HW Hotel Non standard Written Confirmation
Free flow (remark content) and Business Code (3 for Hotel), Language (usually not used), Source (M if Manually added) , Encoding (1 for ASCII)
- **Saving Amount Info**
Qualifier = SAV, Amount and Currency
- **Written Confirmation Contact Info**

Party Qualifier, Communication address identifier (FAX or EMAIL address) and Medium type code

- **Written Confirmation Name and Address Info**

Party Qualifier, Name, address description (2 free flow lines)

- **Arrival Flight Details**

Airline carrier code, flight number, departure city code, Arrival City, Arrival Terminal, Flight Schedule Arrival Time

Manual Auxiliary Segment

Warning: Manual Auxiliary Segments are not available for message versions under 6.2b.

- **ATX Air Taxi Segment**
Carrier code, status code, number in party, board point, off point, requested date, past indicator, free flow text
- **CAR Manual car segment**
Carrier code, status code, number of cars, pick-up point city, pick-up date, drop-off date, car type, past indicator, free flow text, confirmation number
- **HTL Manual hotel segment**
Carrier code, status code, number of rooms, Check-in city, check-in date, check-out date, past indicator, free flow text, accounting information, supplementary information
- **MIS Miscellaneous segment**
Carrier code, status code, number of services required, city code, date of service required, past indicator, service, free flow text
- **TUR Manual tour segment**
Carrier code, status code, number in party, tour start city, tour start date, free flow text, past indicator

Miscellaneous documents

Warning: MCO and XSB are not available for message versions under 6.2b.

Warning: SVC is not available for message versions under 7.1.

The following fares sub-element can be associated to Miscellaneous documents: FA-FB-FE-FG-FH-FI-FM-FN-FO-FP-FS-FT-FZ

- **MCO**
Airline, provider, location, date of service, waiving fee, common fares, type of service, accounting data, issuance reason code, additional passengers, ticket number, international indicator, additional remarks, banker rate, specific fares, fare amount, currency, specific taxes, MCO number in TQM, point of sell, infant indicator.
- **XSB**
Airline, place of issuance, date of service, waiving fee, total amount, charge reason, accounting data, city code from, city code to, number of weight units, value, ticket number, XSB rate, amount, currency, global indicator, pricing airline, XSB pricing routing (from , to, routing carrier, connection point), XSB number in TQM, point of sell.

- **SVC**
Airline, location of the service, date of service, common fares, type of service, accounting data, issuance reason code, additional passengers, ticket number, international indicator, additional remarks, banker rate, specific fares, fare amount, currency, specific taxes, SVC number in TQM, point of sell.

Contact

AP - Contact Telephone Number

This includes the elements with a suffix such as:
AP NCE0493671129-H.

APE - E-mail Address

This is the AP element with the secondary action code E (APE).
Ex: APE-TOMLEE@AMADEUS.NET

APF - Fax Number

This is the AP element with the secondary action code F (APF).
Ex: APF-FRA0123456789

APM – Mobile phone number

This is the AP element with the secondary action code M (APM).
Ex: APM-NCE0612345678

APH - Home Telephone Number

This is the AP element with the secondary action code H (APH).
Ex: APH-FRA69686869

APB - Business Telephone Number

This is the AP element with the secondary action code B (APB).
Ex: APB-FRA45555855

API - Information Telephone Number

This is the AP element with the secondary action code I (API).
Ex: API-PAR456754

APA - Contact Agency phone number

This is the AP element with the secondary action code A (APA).
Ex: APA-PAR456754

AB - Billing Address

It could be either in free-flow text format or in structured format.
In case of free-flow text format, the free flow text format is provided.
In case of structured format, structured information is provided; address options are provided (company name, name, address line 1, address line 2, post-office box, post/zip code, city, state, country).

AM - Mailing Address

It could be either in free-flow text format or in structured format.
In case of free-flow text format, the free flow text format is provided.
In case of structured format, structured information is provided; address options are provided (company name, name, address line 1, address line 2, post-office box, post/zip code, city, state, country).

Moreover, the address type is included if present in the PNR (home, delivery or miscellaneous address).

AQ - Address for credit card

Address used for credit card validation

AI - Accounting information

Data fields could be account number, cost center, client reference, IATA company number.

PNR Security

ES - Extended security¹⁴

Individual PNR security element. (Receiver type, type of access, receiver, creation/update date, originator's details)

Ticketing

TST - Transitional Stored Ticket

Here is a list of the main information from the TST:

- International Sales Indicator
- TST creation date
- Auto/manual indicator
- Last TKT date
- Stopover/Connection indicator
- Origin/Destination
- Fare basis
- Ticket designator
- Not-valid-before and not-valid-after dates
- Baggage allowance
- Fare including currency code
- Equivalent fare including currency code
- Tax including currency code, country tax code and tax type code
- Net fare (Net remit ticketing)
- Fare calculation
- Banker selling rate
- Additional Banker selling rate
- Payment restrictions
- Ticketing carrier (stored as well in the PNR)
- Commission(stored as well in the PNR)
- Tour code(stored as well in the PNR)
- Form of payment(stored as well in the PNR)
- Endorsement(stored as well in the PNR)

¹⁴ Please note that ES element cannot be a triggering element.

- Original issue/in exchange(stored as well in the PNR)

TK - Ticketing arrangement element

The related information provided is:

- Ticketing action identifier: OK, AT, TL, XL...
- Ticketing date/time
- Office ID of the ticket issuance
- Delivering system identification/name
- Transaction information
- Sequence number
- Passenger type: applicable to indicate TK element is specifically associated to an infant.
- Free flow text
- Queue and category number in case of alternative queue placement

FA - Automated ticket identifier

Automatically produced ticket number. The related information provided is:

- Passenger type
- Airline code
- Ticket number
- Ticket status
- E-Ticket carrier
- E-ticket confirmation number
- Currency and total ticket amount
- Ticketing issue date
- Ticketing issue office
- Office ID/IATA that issued the ticket
- Coupon number
- Coupon segment association

FB - AIR Sequence number

The related information provided is:

- Passenger type identifier
- 10 digit A.I.R sequence number
- Entry used to generate the interface record

FD - Fare discount element

The related information provided is:

- Discount code(s)
- Discount percentage

- Status code
- Date of birth/Age
- Airline code
- Numeric value

FE - Endorsement/Restrictions elements

Restrictions, airline comments or rules for TKT. The related information provided is:

- Passenger type
- Airline discount certificate
- Voucher indicator
- Airline code
- Certificate number/Check digit

FG - Shadow AIR sequence number

The related information provided is:

- Passenger type identifier
- 10 digit A.I.R sequence number
- Target office ID

FH - Manual ticket identifier

Used to store manual TKT number. Related information is:

- FH type indicator (Automated ticket number - A, manual ticket number - M, electronic ticket number - E)
- Passenger type indicator (PAX or INF)
- Airline code
- Ticket number / Check digit
- Last conjunction ticket number / check digit

FI - Automated invoice number

The related information provided is:

- Passenger type
- 10 digit sequence number
- Invoice identifier
- Ten digit invoice number

FK - Shadow destination

Office to receive the shadow AIR document. The related information provided is:

- Destination office
- Additional destination office (if present)

FM - Commission

Agent commission basis. The related information provided is:

- Percentage or amount of commission
- Commission indicator

FN - Transmission control number

The related information provided is:

- Passenger type identifier
- Amadeus system provider code
- 11 digit transmission control number

FO - Original issue

Original document number in case of change. Some of the related information provided is:

- Passenger type identifier (if present)
- 3 digit airline code of the original issue document
- 10 digit document number of the original issue document
- Check digit of the original issue document (if present)
- Coupon identifier
- One to four coupons entered in ascending order
- Etc...

FP - Form of Payment

Form of payment details (cash, credit card...), currency, amount.

It may contain a credit card details. The information is transmitted as free-flow text.

FS - Miscellaneous ticketing information

Some of the related information provided could be:

- Free flow text
- Best fare elements for specific airlines
- Etc...

FT - Tour code

Used for negotiated fare on tour rate. Related information is:

- Passenger type indicator (PAX or INF)
- Tour type (IT or BT)
- Last digit of year in which tour code become effective
- Sponsor airline code
- Conference area approval code
- Tour identification
- Net remit indicator
- Staff type

- Staff number

FV - Validating carrier

Used to identify the ticketing airline. Following information is included:

- Passenger type identifier (PAX or INF)
- Validating carrier code (ex: AF)
- Validating office ID for partner ticketing
- Printing options

FY - Fare print override

Some of the related information provided could be:

- Passenger type identifier
- Base fare override
- Equivalent fare override
- Tax amount override
- Total fare override
- Etc...

FZ - Miscellaneous ticketing information 2

It differs from the FS element in the sense that it is not linked to the TST. Some of the related information provided could be:

- Free flow text
- Best fare elements for specific airlines
- Etc...

Options

OP - Option Element

Place PNR on a specific queue at specified time/date.

OPQ - Confidential option element

Place PNR on a specific queue at specified time/date.

OPX - Confidential cancellation element

Cancel itinerary if no TKT nos. present at specified time/date.

OPC – Automatic cancellation element

Cancel the eligible segment(s) if no TKT present at specified time/date. The OPC rule ID (also known as ABR rule ID) that has created the OPC element as well as the cancellation date can be included.

OPW – Automatic Warning element

Place the PNR on a specific queue at specified time/date if no TKT is present at specified time/date.

SSR - Special Service Request

All the SSRs could be provided.

Each of the SSR codes that have to be sent must be selected explicitly.

Following information is included:

- Requirement type (SSR code)
- Airline code
- Status code (if present)
- Number in party
- Free-flow text

Frequent flyer SSRs (SSR FQTx)

In addition to the standard SSR details, following information is included:

- Frequent traveler description (company, FF number, tier level, priority number, airline tier level, customer value, alliance priority code, alliance tier description, alliance code)
- **Warning:** Redemption and Upgrade details are not available for message versions under 6.1.
Frequent traveler verification for redemption and upgrade (action request, FF number, rate/tariff class identification, sequence number, reference version number, rate/tariff class, approval identification)

Seat Request (SSR SEAT)

In addition to the standard SSR details, following information is included:

- Date of change of gauge
- Boarding-pass indicator for change of gauge
- Area of preference
- Seat characteristic
- Smoking/Non smoking
- Seat number
- Seat row

Group Seat Request (SSR GPST)

In addition to the standard SSR details, following information is included:

- Number of seats
- Smoking/Non smoking
- Seat range(s)

SK - Keyword

This is the SK element.

Each of the keyword codes that have to be sent must be selected explicitly.

The following information is sent in case the access to this element is not restricted for the datafeed office.

- Keyword code
- Airline code
- Status code (if present)
- Number in party (if present)
- Free flow text (if present)

This element is a secured element; therefore its access is dependent on the access rights of the datafeed process. These access rights are inherited from an office ID which is chosen for each airline-authority couple. If this office does not have access to the SK element, the element will not be sent at all (on a PNR display, the SK element would be shown as "Restricted").

Remarks

RM - General Remarks

Information transmitted is:

- Remark category
- Free flow text

RI - Document remarks

Displayed on invoice/itinerary documents. Information transmitted is:

- Remark category (A, B, C, D, F, I, M, O, P, R, S, T, U, Z)
- Free flow text

RC - Confidential remarks

It contains the free text of the remark and the security elements

This element is a secured element; therefore its access is dependent on the access rights of the data feed process. These access rights are inherited from an office ID which is chosen for each airline-authority couple. If this office does not have access to the RC element, it will not be sent at all (on a PNR display, the RC element would be shown as "Restricted").

RX - Corporate remarks

Information transmitted is:

- Free flow text

This element is a secured element; therefore its access is dependent on the access rights of the data feed process. These access rights are inherited from an office ID which is chosen for each airline-authority couple. If this office does not have access to the RX element, it will not be sent at all (on a PNR display, the RX element would be shown as "Restricted").

RQ - Quality control remarks

Used by some front offices quality control application. Information transmitted is:

- Free flow text

OSI - Other service

Following information is transmitted

- Airline code
- Free-flow text

Split and Cross-Reference¹⁵

SP - Split information

This is the SP element of split PNRs (present in both parent and associated PNR). In case of split PNRs, the SP element is sent (providing the date of the split, the office/agent identification and the record locator of the parent or associated PNR); the split direction is sent as well in the way that for each record locator an indicator tells if this is a parent or a child of this PNR.

The information transmitted is:

- Date of the split
- Sign of the agent who performed the split (agent initials and duty code)
- Office ID where the split was made
- Record locator and creation date of the associated PNR

Warning: Creation Date is not available for message versions under 7.1.

- Split direction indicator (Parent, Child or Non Homogeneous indicator)

AXR - Associated cross reference record

The AXR shows the record locators of all passengers who were originally in a same PNR but have now been split into multiple PNRs

The information transmitted is:

- Date of the action (split, replication...)
- Record locator of the associated PNR
- First 12 characters stored in the AXR record

RR - RR

A PNR is copied with an AXR creation; this RR element contains the copy date, the initials and duty code of the agent, office ID, origin record locator.

The information transmitted is:

- First date

¹⁵ Please note that cross references elements cannot be a triggering element.

- Sign of the agent who performed the replication (agent initials and duty code)
- Office ID where the replication was made
- Record locator of the associated PNR

PDI

Warning: PDI are not available for message versions under 5.1.

Altea DepartureControl directly updates PNRs with PDI data and its PDI data can be part of the feed.

But some other DCS (Departure Control System) are linked to Amadeus system and can send back to Amadeus changes made during DCS time (PNR modifications or creation). PDI data added by these DCS are taken into account on Amadeus PNR and can then be part of data feed.

The information transmitted is:

- Board point, Off point
- Leg number
- Seat delivered details (jump seat supported)

Warning: Jump Seat is not available for message versions under 7.1.

- Cabin identification
- Acceptation/Regrade
- Customer Status
- Inbound connection flight details (Departure date/time, Arrival date/time, flight number, board/off points, booking class) and booking status.

Warning: Inbound connection flight details are not available for message versions under 8.1

- Outbound connection flight details (Departure date/time, Arrival date/time, flight number, board/off points, booking class) and booking status.

Warning: Outbound connection flight details are not available for message versions under 8.1

- Billing information: Check-in airline, IATCI originator, IATCI Target DCS, IATCI Target company, Onward flight carrier, Onward flight number, Onward flight suffix, Onward flight departure date

Warning: Billing information is not available for message versions under 7.3.

- Acceptance channel information (acceptance origine, application type) – Billing information

Warning: Acceptance channel information is not available for message versions under 8.1.

- Baggage details (number of checked baggage, checked baggage weight, weight indicator, unchecked weight, baggage tag numbers, Final destination to which bags are checked, baggage pooling reference number)

Warning: Baggage information is not available for message versions under 8.1.

- Amount of Compensation (Type (Voluntary or Involuntary) and Free-flow text containing amount of compensation)

Warning: Amount of Compensation is not available for message versions under 8.1.

DID

Warning: DID are not available for message versions under 7.3.

DID is the SBR Identification element. This is a technical unique identifier for a passenger/segment. This information is not displayable on the PNR.

CUS

Warning: CUS are not available for message versions under 8.1.

A customer profile is a data record stored and maintained in the AMADEUS Central System. The system user may create the profile to contain any information about the companies or travellers who make regular bookings in order to assist with the reservation process.

This can include information on traveller's preferences, frequent flyer details, seating requirements, and other information such as company corporate travel policies and payment details. An agency can also have its own detail in a profile to facilitate the booking.

The information transmitted the PNR data feed is:

- The profile type (Traveller, Frequent flyer, Company, Corporate, Agency, Staff).
- The record locator of the profile.

Annex 2: PNR Data elements choice and granularity

The table below indicates the elements that can be chosen to trigger the feed with upon an activity (Addition/Modification/Deletion/Presence) and those that can be part of the output.

Element family	Acronym	Can trigger the feed on the following activity				Activity in Output	Can be present in Output
		Addition	Modification	Deletion	Presence		
<i>PNR Header¹⁶ (With PoS of Owner, Creator, Last Updator)</i>		N	Y	N	N	Y	Always
<i>Air Itinerary (With PoS at segment level)</i>							
	Air segments	Y	Y	Y	Y	Y	Y
	Informational segments	Y	Y	Y	Y	Y	Y
	Open segments	Y	Y	Y	Y	Y	Y
	ARNK segments	Y	Y	Y	Y	Y	Y
<i>Non Air Itinerary</i>							
	Car segments	Y	Y	Y	Y	Y	Y
	Hotel segments	Y	Y	Y	Y	Y	Y
<i>Manual Auxiliary Segments</i>							
	ATX Air Taxi	Y	Y	Y	Y	Y	Y

¹⁶ The only activity reported in case of change in the header of the PNR is a change in the RP line (owner office field or queuing office field). Hence changes in the PoS of the last updator cannot be a triggering element.

Element family	Acronym	Can trigger the feed on the following activity				Activity in Output	Can be present in Output
		Addition	Modification	Deletion	Presence		
	CAR Manual car segment	Y	Y	Y	Y	Y	Y
	HTL Manual hotel segment	Y	Y	Y	Y	Y	Y
	MIS Miscellaneous segment	Y	Y	Y	Y	Y	Y
	TUR Manual tour segment	Y	Y	Y	Y	Y	Y
Miscellaneous information documents ¹⁷							
	MCO	Y	Y	Y	Y	Y ¹⁸	Y
	XSB	Y	Y	Y	Y	Y ¹⁹	Y
	SVC	Y	Y	Y	Y	Y ²⁰	Y
DCS							
	PDI Data	Y	Y	Y	Y	Y	Y
Name							
	NM	Y	Y	Y	Y	Y	Y
	NG	Y	Y	Y	Y	Y	Y
Contact							
	AP	Y	Y	Y	Y	Y	Y
	APE	Y	Y	Y	Y	Y	Y
	APF	Y	Y	Y	Y	Y	Y
	APH	Y	Y	Y	Y	Y	Y
	APB	Y	Y	Y	Y	Y	Y
	API	Y	Y	Y	Y	Y	Y

¹⁷ To have Fare elements associated to Miscellaneous Information Documents as trigger/output, Fare elements have to be chosen in the trigger/output.

¹⁸ The old value is not available in case of modification or deletion

¹⁹ The old value is not available in case of modification or deletion

²⁰ The old value is not available in case of modification or deletion

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Functional Marketing Specification**

Element family	Acronym	Can trigger the feed on the following activity				Activity in Output	Can be present in Output
		Addition	Modification	Deletion	Presence		
	APM	Y	Y	Y	Y	Y	Y
	APA	Y	Y	Y	Y	Y	Y
	AB	Y	Y	Y	Y	Y	Y
	AM	Y	Y	Y	Y	Y	Y
	AQ	Y	Y	Y	Y	Y	Y
	AI	Y	Y	Y	Y	Y	Y
<i>PNR Security</i>							
	ES	Y	Y	Y	Y	Y	Y
<i>Ticketing</i>							
	TK	Y	Y	Y	Y	Y	Y
	TST	Y	Y	Y	Y	Y	Y
	FA	Y	Y	Y	Y	Y	Y
	FB	Y	Y	Y	Y	Y	Y
	FD	Y	Y	Y	Y	Y	Y
	FE	Y	Y	Y	Y	Y	Y
	FG	Y	Y	Y	Y	Y	Y
	FH	Y	Y	Y	Y	Y	Y
	FI	Y	Y	Y	Y	Y	Y
	FK	Y	Y	Y	Y	Y	Y
	FM	Y	Y	Y	Y	Y	Y
	FN	Y	Y	Y	Y	Y	Y
	FO	Y	Y	Y	Y	Y	Y
	FP	Y	Y	Y	Y	Y	Y
	FS	Y	Y	Y	Y	Y	Y
	FT	Y	Y	Y	Y	Y	Y
	FV	Y	Y	Y	Y	Y	Y
	FY	Y	Y	Y	Y	Y	Y
	FZ	Y	Y	Y	Y	Y	Y
<i>Options</i>							
	OP	Y	Y	Y	Y	Y	Y
	OPQ	Y	Y	Y	Y	Y	Y
	OPC	Y	Y	Y	Y	Y	Y
	OPW	Y	Y	Y	Y	Y	Y
	OPX	Y	Y	Y	Y	Y	Y

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Element family	Acronym	Can trigger the feed on the following activity				Activity in Output	Can be present in Output
		Addition	Modification	Deletion	Presence		
<i>Seats</i>							
	Seats	Y	Y	Y	Y	Y	Y
<i>OSI</i>							
	OS	Y	Y	Y	Y	Y	Y
<i>SSR</i>							
	SR	Y	Y	Y	Y	Y	Y
	SR+CODE	Y	Y	Y	Y	Y	Y
	SR OTHS/ADTK 1A/ZZ	Y	Y	Y	Y	Y	Y
<i>Keywords</i>							
	SK	Y	Y	Y	Y	Y	Y
	SK+CODE	Y	Y	Y	Y	Y	Y
<i>Remarks</i>							
	RM	Y	Y	Y	Y	Y	Y
	RC	Y	Y	Y	Y	Y	Y
	RX	Y	Y	Y	Y	Y	Y
	RI	Y	Y	Y	Y	Y	Y
	RQ	Y	Y	Y	Y	Y	Y
<i>Cross references</i>							
	SP/Split	N	N	N	Y	Y ²¹	Y
	RR/Replication	N	N	N	Y	Y ²²	Y
<i>Technical Data</i>							
	DID	N/A	N/A	N/A	N/A	Y	Y

²¹ The activity available is not the activity on the SP element, but the activity on the split process. If activity on split process is selected, SP element must be selected in the Output.

²² The activity available is not the activity on the RR element, but the activity on the replication process.

Element family	Acronym	Can trigger the feed on the following activity				Activity in Output	Can be present in Output
		Addition	Modification	Deletion	Presence		
<i>Customer Profile</i>							
	CUS	Y	Y	Y	Y	Y	Y

Annex 3: Product Policy

The data feed is a community product. This means that all airlines can benefit from developments requested by other airlines. However, offering a single product for everyone means Amadeus must set certain policies governing the feed, to assure a proper balance between product evolution and technical compatibility. These rules are described in this appendix.

Terms

Major upgrade

A change to the feed that requires a change in grammar structure, and therefore requires development at both ends to support the new format.

New Release

A backward compatible change to the data feed that requires only a minor adjustment to the feed, to receive the enhancement. Should an airline decide not to make this change, it does not affect the feed it receives. A technical document is available upon demand that explains the notion of compatibility in more detail with examples.

Policy

- 1) Amadeus will issue no more than three product releases per rolling 12-month period; one may be a major upgrade. The aim is to limit the number of times the feed is changed (to limit customer disruption), while also allowing the product to evolve and improve.
- 2) Amadeus will ensure no fewer than three months between releases. This limits the frequency with which the feed is adjusted, and the amount of upgrade work needed. Example: Airline 6X is on V1.0. Airline 7X needs an element added to the feed that slightly affects messaging. Amadeus therefore releases V1.1, but if 6X does not want/need this enhanced field, it can stay on V1.0. and still read the feed.
- 3) Three concurrent versions of message releases will exist. Four versions are supported for a period of three months to give customers time to manage a software upgrade to a newer version. As explained above, this means that Amadeus does not ask customers to immediately upgrade each time a feed changes. Instead, it enables customers to keep their existing feed in place for a minimum of nine months. Most importantly, it implies a three-month notice period before they need to upgrade. If a customer requests a feed change, they are obliged to upgrade to the latest version.
- 4) The above applies to both EDIFACT and XML feeds, and batch and online feeds.
- 5) Customers can receive data elements present in the current version of the message, and can benefit from new elements that are added as the product evolves, provided they upgrade to the new message version.
- 6) Customers can change their triggering conditions and output received (PNR view) up to two times per year, free of charge. Additional changes may be made, and the charge for this extra work is evaluated on a case-by-case basis.

- 7) Customers will receive data directly related only to their business, even if they are able to view complimentary PNR elements in the system. This is typically defined by the ownership of the PNR, or by the marketing carrier responsible for particular travel segments. In practice, it means a customer will receive the complete itinerary, but not the auxiliary information related to a passenger travelling on those segments. For example: Airline 6X will not see Special Service Requests (SSRs) related to the 7X leg in the same PNR, even if it is a connecting incoming flight, and even if 6X can see the itinerary and SSRs for the YY in the system.
- 8) Amadeus will provide the customer with a suitable test feed for a period of four weeks prior to planned go-live, and for two weeks after go-live.
- 9) Two initialisation uploads will be provided as part of an implementation or a migration to Amadeus Altéa Reservation.
- 10) Should customers wish to receive the test feed for more than four weeks (as outlined in point #8 above), or wish to receive more initialisation uploads than specified (see point #9 above), these services must be pre-arranged. This arrangement would become an Amadeus project migration or implementation, and would be subject to charges determined on a case-by-case basis.
- 11) Transmitted data is kept for a maximum period of three days by Amadeus to support clients in the event of a problem with the integration or on the client side.
- 12) The Implementation time-frame for PNR Data Feed is 6-8 weeks, after the kick off and reception of all the required information for the implementation, which include filled implementation questionnaire, FTP/MQ Contact, Office ID, etc.